JOINT COMMITTEE ON RHEOLOGY
OF THE
INTERNATIONAL CONGRESS OF SCIENTIFIC UNIONS

REPORT
ON THE PRINCIPLES OF RHEOLOGICAL NOMENCLATURE

by

J.M. BURGERS and G.W. SCOTT BLAIR

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PREFACE.

In its first meeting, September 8-9, 1947, held in London, the Joint Committee on Rheology decided to take up the subject of nomenclature as a suitable theme for promoting international contact between rheologists. Two members of the Committee, J.M. Burgers and G.W. Scott Blair, were charged to prepare a Report on this subject. As a starting point they could make use of a paper on the "Nomenclature of Deformations", published originally in 1940/1941 in the Dutch language by the Committee for the Study of Viscosity and Plasticity of the Royal Netherlands Academy of Sciences at Amsterdam, and of a series of proposals brought forward by the British Rheologists' Club and partly published in "Nature" 149, p.702, 1942. Some discussion of these papers which in several respects showed a remarkable parallelism, took place at the meeting of the Committee.

A first draft for a report on nomenclature was written by the authors in the beginning of 1948. After having been printed, it was forwarded to a great number of scientists interested in the subject, so that it could serve as a basis for a discussion at the International Rheological Congress held at Scheveningen (Holland), September 20-24,1948. Many speakers took part in the discussion and several of them have afterwards submitted written contributions in which they explained their points of view.

The Joint Committee, which met during the Congress, decided that a new Report should be prepared by the same authors, in which regard should be taken of the result of the discussions. This revised and amplified Report would be published in the Proceedings of the International Rheological Congress, while a sufficient number of separate copies should be available to the Committee for distribution. It is given in the following pages. The Joint Committee on Rheology does not intend to take decisions concerning nomenclature. What the Report brings are recommendations, based on an analysis of phenomena of deformation, mainly such as are produced under the action of a constant shear stress. It is intended to hold the third meeting of the Committee at Amsterdam, in September 1949, in close connection with the General Assembly of the International Union of Chemistry, which Assembly will bring together several Committees of the IUC interested in the subject of deformations and their nomenclature. This will afford an opportunity for further discussion and, it is hoped, for obtaining support for the Committee's recommendations.

In preparing the revised Report the authors have extended the treatment of the basic phenomena, in order to give greater clearness to the distinctions introduced. In view of the time available this made it necessary, on the other hand, to limit the scope of the subject and to restrict the phenomenological descriptions to the case of shear. It is assumed, either that the deformation of the material is a simple shear, in which case attention is only given to the corresponding shear component of the resulting stress system; or that the material is subjected to a pure shear stress test, in which case attention is given to the corresponding shearing deformation. Other

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components of stress which might appear in the first case, or extensions and contractions which might accompany the shearing deformation in the second case, are not considered.

The main divisions proposed in the Report are based on experiments in which the material, for a period of given duration, is supposed to be The main divisions proposed in the Report are based on experiments subjected to a shear stress of constant magnitude, which at the end of that period is taken away. The magnitude of the shear is observed as a function of the time, both during the period of action of the stress and after its removal. Experiments in which the magnitude of the deformation is taken as an independent variable, while the course of the ensuing stress is observed as a function of the time, have briefly been consider- ed in a separate chapter (Chapter IV). Experiments in which the material is subjected to a periodically varying stress and a periodically varying deformation, have been touched upon in section 48. In view of the special nature of this subject, however, it was thought better not to enter into questions of its nomenclature, which should be treated separately at some
other occasion.

The results of experiments on the relation between tensile stress and extension will in general show analogies with those obtained from experiments on shear. Hence a nomenclature based on the outcome of such experiments must, on the whole, be similar to the one developed here. Nevertheless, there will appear certain differences connected with the geometrical character of the deformation. A treatment of the analogies and differences would require the consideration of general spatial stress systems. It was thought that this subject still presents many difficulties and it has been left aside in the present Report. Likewise the problems presented by large deformations and such phenomena as are expressed by the provisional terms "climbing phenomena", "Spinnbarkeit", etc. have been left aside. The same applies to phenomena of rupture and to "turbulence", whether this is true turbulence of the type considered in hydrodynamics or embraces phenomena of a particular kind, to be observed with certain non-Newtonian liquids or other special fluids. It is hoped that some of these subjects can be treated at a further stage of the Committee's work.

Likewise optical, electrical and magnetical phenomena which may accompany deformation or flow, have been omitted.

Concerning the dependence of rheological phenomena on temperature, it has been assumed that observations and sets of observations such as e.g. are required for the construction of a flow-diagram (compare section 19), are carried out at a constant temperature. It has been indicated in various places that a more complete characterisation of the rheo- logical properties of a substance can be obtained by performing series of observations at different temperatures. However, in the present Report terminology has been developed to characterise particular forms of temperature dependence. Also this subject can be discussed more properly at a later stage. For the same reason thermodynamical considerations have been left aside, the more so as practically all rheological phenomena are accompanied by irreversible processes in the thermodynamic sense, which make an exact treatment of this subject particularly difficult.

Although the subject of the rheology of liquid suspensions likewise has been considered as falling outside the scope of the present Report, some recommendations concerning terminology to be used in characterising the viscosity of suspensions have been given in the Appendix.

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As regards discussional observations, written remarks had been sent in by Mr. J. Pryce-Jones, Sir Charles Goodeve, Professor M. Reiner, Dr. J.G. Oldroyd and Dr. R. Houwink. The rather extensive observations by Pryce- Jones and by Goodeve which were based partly on publications by these authors, have been treated at some length in sections 5252 - 53 and 56 - 57, of Chapter VII, in which Chapter a comparison is given between the system of nomenclature recommended in the present Report, and proposals made by other authors.

Prof. Reiner pointed out that one cannot rigorously define real things but that one must be content to define concepts which can form the basis for mathematical treatment. By way of example he mentioned that one cannot say either "steel is a solid" or "steel is a liquid", as every material can show various forms of rheological behaviour, depending on the circumstances to which it is submitted. Prof. Reiner contends that one can only say: a Hookean body is a solid" or "a Newtonian material is a liquid", where "Hookean body" and "Newtonian liquid" are abstract concepts. The authors have attempted to evade the difficulty pointed out by Prof. Reiner as far as possible, by defining not bodies, but types of rheological behaviour. They believe, nevertheless, that in certain cases one can define a type of body (e.g. a solid), as presenting a certain form of behaviour under specified circumstances. It is further to be kept in mind that, whenever such definitions occur in the present Report, it is understood that the definition is valid only at the temperature at which the analysis of the behaviour has been carried out. Reiner's standard types of rheological behaviour have been briefly enumerated in section 63.

Dr. Oldroyd drew attention to the extreme complexity of the problem of classifying actual materials by their rheological properties, as it will usually be impossible to specify the properties of a material completely by means of differential equations of state. And even when it is assumed possible for an actual material, such equations must in general be supposed to take the form of functional relationships between an infinite number of variables. The authors would repeat that they attempted to evade this difficulty by specifying forms of behaviour rather than aiming at a full description of the properties of types of material. But again they would contend that a classification of materials of sufficient practical importance is possible by restricting oneself to such aspects of their behaviour, as become evident from experiments of a single type or of a few types, leaving it to more refined research to make further distinctions.

Dr. Houwink stressed the importance of the time-scale in defining concepts as yield-value and the like. In particular he considered the "rate of application of the stress", expressed by $d\tau/dt$, as a quantity of importance, and mentioned that water seems to behave as a solid, when a flat stone is thrown along a water surface at a small angle, in which case the stone rebounds. The authors would point out that in this case inertia effects are determining the observed behaviour; in drawing up the definitions of their Report, they intended to exclude such effects (section 4). Granted the possibility of excluding inertia effects, it is assumed that in the standard type of experiment the shear stress is brought to its final magnitude in such a short interval of time, that practically no deformation of the material has occurred before the maximum value is reached. It is therefore impossible to insert a clause into the definitions concerning the value of $d\tau/dt$, unless one were to tackle the subject of inertial effects on rheological phenomena, which. would require a special study of great length. On the other hand, due attention has been given in the Report to the importance of the time- scale applied in observing the course of a deformation as a function of time.

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Dr. H. Leaderman, of the National Bureau of Standards, Washington D.C., had the kindness to forward a copy of a paper on "Rheological Nomenclature for Elastic and Inelastic Behaviour of High Polymers", presented at the meeting of the (American) Society of Rheology, New York, November 6, 1948. Reference to some of the terms applied by Dr. Leaderman has been given in sections 58 and 59.

Attention has further been given to views on thixotropy and related forms of rheological behaviour developed by the late Dr. H. Green and Miss R. M. Weltmann (sections 54-55); to the concept of "creep" as it occurs in some papers (section 60); to a few terms used by W. Philippoff in his book "Die Viskosität der Kolloide", Dresden und Leipzig 1942 (section 61); and to Nutting's equation and related expressions (section 62).

In order to help the reader to follow the main distinctive points introduced in the Report, a Table has been prepared giving a summary of the principal headings. Through the courtesy of Dr. E. W. J. Mardles, Secretary of the British Rheologists' Club, a provisional form of this Table was forwarded to a number of rheologists: when printing the Report it was possible to take notice of their observations. The authors wish to express their great thanks to Dr. Mardles, and likewise to all persons who in one way or another have contributed to making their work more complete.

It had been decided at the 1948 meeting of the Joint Committee on Rheology to make available a translation of the main terms of the Report in the French and the German languages. Time has not yet permitted us to execute this plan. It is hoped that translations into French and German, and also an Esperanto translation, of the Table accompanying the Report may be ready before the meeting of September 1949.

Delft/Reading
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Part A: GENERAL TERMS

Chapter I – Preliminary considerations

1. The development of a system of nomenclature in rheology has to satisfy two needs:
   (1) The classification of types of rheological behaviour;
   (2) The classification of materials according to their rheological behaviour under specified conditions.

All the concepts to be introduced must be based on well established mechanical and physical principles. At the same time it is evident that a classification of types must have regard to experimental practice and to observed forms of behaviour. References to phenomena exhibited by classes of substances will therefore frequently turn up in analysing types of behaviour.

In describing rheological behaviour we shall restrict ourselves to those phenomena where the behaviour of an element of volume is representative of the material. No attention is to be paid to peculiarities effects considered in this Report. This implies that we shall restrict ourselves to the consideration of "homogeneous" materials subjected to a homogeneous (or "uniform") field of stress. We assume that under these conditions the deformation of the material likewise will be homogeneous (uniform).

The terms "homogeneous" and "uniform" will be defined from a macroscopic point of view, the scale being fixed by the magnitude of the elements of volume or of area over which a macroscopically homogeneous stress can be applied. If a material is not homogeneous microscopically, it must everywhere present the same average composition and structure (including orientation of structure, if such orientation is present) The averages are calculated with reference to elements of average stress components, and are thus typical macroscopic entities, subject to the condition that the average stress components shall not appreciably vary over distances of the same order of magnitude.

When the structure of the material is anisotropic (this term is not restricted to anisotropy which can be detected by optical methods, but refers to any property which appears to be connected with definite directions in the material), the orientation of the structure must not vary over distances of the order of magnitude just mentioned. The relative orientations of the structure and of the applied stress must be specified.

2. The particular difficulty of the subject is due to the circumstance that most materials suffer irreversible internal changes when subjected even to simple types of deformation. Moreover, those entities which we call "physical properties" are almost invariably changed reversibly by the testing process even when no irreversible change takes place. This makes it necessary in many instances to base a classification of materials on an analysis of their behaviour during a specified schedule of operations, rather than on "objective" properties which can be assigned to the material as such. Along with this it becomes necessary to give due attention to time effects in specifying types of rheological behaviour. As the possibilities in this direction can be extended indefinitely, a limitation to the number of cases to be considered must be introduced, as otherwise the subject will become intractable.

The most direct evidence of irreversible behaviour is found in the appearance of non-recoverable or permanent changes of form. This phenomenon is of fundamental importance and is used as a basic principle in developing classification

It should be noted that the permanent or non-permanent character of a change of form in general is not discernible so long as the deformation is in progress, but must be inferred from observations after the process has stopped.

3. The appearance of a permanent change of form, although proof of an irreversible change of position of the molecules or atoms out of which a substance is built up, need non imply change of internal structure such as would lead to a change of physical properties. The observable physical properties of a substance are dependent on the statistical character of its molecular (or atomic) structure, and it is possible that this statistical character is not changes, although the positions of the individual molecules have been altered.

Nevertheless, so long as change of form is in progress, the statistical character of the molecular (or atomic) arrangement can deviate from the normal one. There may appear a spontaneous return to the original statistical character when the substance, after the experiment, is left to itself. This phenomenon will be called "re-adjustment". It is characterised by a time-scale (average time of re-adjustment), which for certain substances (i.e. ordinary liquids) appears to be extremely small, whereas for other substances it can be large. Closer investigation may reveal that the description of its course as a function of time needs the introduction of a whole spectrum of re-adjustment times. When this spectrum contains infinite re-adjustment times, the re-adjustment will be incomplete.

Re-adjustment of molecular structure can already start during the progress of a change of form. It has its influence on retardation phenomena (section 32). Another consequence of re-adjustment is the phenomenon of stress relaxation (section 46).

4. It will be supposed that the state of stress to which the material is subjected, can, when necessary, be kept unchanged during a certain length of time. Of particular interest, are cases where a steady state is obtained; either a state in which the deformation of the material has assumed a form independent of time, or a state in which the deformation changes at a uniform rate. It is possible that the steady state is reached after a certain lapse of time, or that it is approached only asymptotically.

Cases where the deformation or the rate of deformation changes appreciably with time, must be fully specified as such.

Considering experiments in which the material is subjected to a constant stress, it will be assumed that the external forces through which the stresses are produced, are applied in such a way that the internal stress increases from zero to its final value in a monotonic way. Cases in which the material temporarily comes under the influence of a stress larger than which is present in the experiment proper, are excluded.

As the physical properties of certain materials change during the process of deformation, it is necessary that the external forces be applied sufficiently quickly to prevent the material from suffering a permanent change of structure (of a magnitude comparable with that produced by the deformations to be investigated), before the force has reached its final value. In practice it is not always possible to prevent this, since at the same time care must be taken that too rapid an application of force does not lead to the appearance of vibrations. The experimenter must consider in what we he can apply the exterior forces so as to fulfil both conditions as well as possible.

In this Report, we shall not enter into the details of this problem and, for simplicity of expression, shall described the application of the forces as instantaneous. Likewise, we shall assume that their instantaneous removal is possible. It must be kept in mind, however, that in certain cases (in particular with forces of great magnitude) it will be necessary to investigate the possible influence of inertial reactions.

When an experiment is carried out, attention must be given to the circumstances that in applying a system of forces to a deformable material, work is performed upon this material. Part of this work can be stored as an increase of internal energy, connected with displacements of atoms: another part will be converted to heat. Thermodynamical considerations point to the effects of the presence or absence of heat exchange with the surrounds on the phenomena of deformation, e.g. on the magnitude of elastic reactions. Experiments should therefore be performed in such a way that the conditions of heat are fully known. The best conditions for exact measurements are obtained when the experiments are performed either adiabatically (without any heat exchange) or isothermally (heat exchange so adjusted as to ensure a constant temperature during the whole experiment including the periods of application and removal of the force). Where these conditions cannot be satisfied, information about the possibilities of heat exchange should be given as far as the experimental arrangement permits.

The importance of the effects of heat exchange are greatest in cases where the force I varied rapidly, e.g. in experiments with oscillating forces. When the experiment is directed to the observation of a steady state, conditions will usually be approximately isothermal. Nevertheless, even in such cases, the conditions present during the period of application of the force may be of importance, while those present during its removal may influence the recovery phenomena.

As rheological properties usually are very sensitive to changes of temperature, the ensurance of isothermal conditions is generally to be preferred.

5. Considering experiments in which the stress has been kept constant over a certain period, it is of importance to observe what happens when the external force acting on the specimen is removed. The following phenomena call for attention.

A) The extent of return to the original shape of the specimen.

The deformation may disappear either inappreciably of partially or completely. If the second or third case presents itself, the material is said to behave partially or completely elastically, and the deformation which existed during the application of the force is called an elastic deformation, to the extent (partial or complete, or to a certain percentage) to which it disappeared on removal of the external force. The non-disappearing part of the deformation is called the permanent deformation. When the deformation disappears completely, we speak of perfectly or completely elastic behaviour.

B) The degree of return to the original physical state and properties of the specimen.

There may be no appreciable return, or a partial or a complete return. The distinction between the first and second possibilities may often be difficult. In most cases only a distinction between a complete return and a permanent change of physical state will be made. In the former case the deformation is said to be not affecting the physical state; in the latter state it affects the physical state.

In many cases it can be accepted that when there is a complete return to the original shape of the specimen, so that its external behaviour appears to be perfectly elastic, there will be

no change in its physical state. Although this must not be taken as a general rule, it will be evident that the distinction sub B) mainly refers to the case of incomplete elastic behaviour or absence of elasticity.

Both the (complete or partial) disappearance of a deformation and the return to the original physical state are designated by the term recovery. There is thus; recovery of shape (e.g. of length), and recovery of physical state.

The recovery often does not happen immediately, but may take a certain time. This is a consequence of the circumstance that all re-adjustment of molecular structure involves certain time-scales.

The recovery sometimes does not happen spontaneously, but appears only after some form of stimulations.

In the present Report, the application of the term "change of physical state" will be restricted to such changes of state as are evident from the rheological behaviour of the specimen when subjected to a renewed test. As rheological properties are usually extremely sensitive to changes of physical state, this restriction will not seriously reduce the number of important cases.

6. Experiments in which a sample of the material during a given period is subjected to a given external force, must in general be carried out for various periods of time and with various magnitudes of the force. It is to be observed that, in the rather frequent case where the properties of the material change in consequence of the experiment, every new experiment must be made on a fresh sample.

It is assumed that we have at our disposal an unlimited supply of the material, in a sufficiently homogeneous and stable condition that a repetition of the same test (same value of the force and same duration of application, each test being applied to a fresh sample) will always give the same result. We shall indicate this by saying that the tests must be replicable.

Repeated tests performed on the same sample can be used to investigate whether the properties of the specimen have or have not changed during the experiment. In the latter case we shall say that the test is reproduceable. Important cases of non-reproducibility are presented by the phenomenon of work-hardening and thixotropy (Chapter V – Application of repeated experiments performed on a single specimen.).

Along with experiments in which the force is kept constant during a certain period and is then removed, it is possible to consider the behaviour of a material subjected to a harmonically oscillating load, either with a zero mean value, or with a finite mean value. Such experiments are applied in investigation on internal damping, hysteresis, and fatigue.

Another set of cases is obtained by prescribing the course of the deformation as a function of time (instead of the time-dependence of the force), e.g. the deformation be kept constant (as in relaxation experiments), or slowly increased at a uniform rate.

Chapter II – Elements of stress and strain analysis.

7. Stress is defined as the internal force in matter, transmitted per unit of area between adjacent

layers. This force is vector, the magnitude and direction of which are functions of the direction of the normal to the element of area considered. The state of stress at any given point consequently forms a tensor of the second rank.

A stress component is reckoned positive, when the material on the positive side of the element of area exerts a traction on the material on the negative size. When the stress component along the normal to the element is positive, the component is called a positive normal stress or tension; when it is negative, it is called a negative normal stress or pressure. Stress components parallel to the element are called (positive or negative) tangential stresses or shear stresses.

Some authors prefer to call the vectorial components of stress "tractions".

With reference to an orthogonal system of coordinates the stress tensor has three normal components and three pairs of shear components.

A state of stress is a phenomenon which refers to a definite instant. Magnitude and direction of the force, and magnitude and position of the element of area are measured in the actual state of the material, so that the stress can be fully described as it is found at that instant, without reference to previous instants.

Besides the internal force, measured by the stress, the material may be subjected to external forces, of which the most common is the force of gravity. When the stresses acting from all sides on the boundary surface of an element of volume, together with the external forces acting on the matter within this boundary surface, are not in equilibrium, the element will suffer a certain acceleration, such that the inertial forces called into play make up the balance.

Although the definition of stress given above is equally applicable to cases where there is equilibrium or where there is no equilibrium, we shall be concerned in the present Report only with cases in which accelerations are absent or are at least so small that the part played by inertial forces is negligible. Also the influence of the external forces will be supposed to be so small that the stress system may be considered as homogeneous.

8. A state of stress is described by a symmetrical tensor of the second order rank:

$$||P||$$

Equation 1

Its components with reference to Cartesian coordinates x, , y, z, can be divided into two groups:

p_{xx}, p_{yy}, p_{zz} (or, for shortness, p_x, p_y, p_z): normal or tensile tractions

$p_{xy} = p_{yx},\ p_{xz} = p_{zx},\ p_{yz} = p_{zy}$: tangential or shear tractions

Equation 2

A homogeneous or uniform state of stress is one for which the components are independent of the coordinates.

A uniform state of stress, independent of the material in which it is present, can always be considered as the resultant of a uniform hydrostatic state of stress, i.e. an isotropic tension or

pressure, and a uniform deviatoric state of stress, characterised by the property that the sum of its three principal components is zero.

The simplest uniform deviatoric state of stress is two-dimensional, in which there are only two principal stress components, of equal and opposite magnitude, the third principal component being zero. When a coordinate system is introduced, the x- and y- axes of which make angles of 45° with the directions of the two principal stress components, while the z-axis is perpendicular to these, a simple shear stress is found in the planes parallel to the x, z-plane (p_{yx}) and in those parallel to the y, z-plane (p_{xy}), being everywhere of the same magnitude for both sets of planes.

9. In a homogeneous isotropic material, a uniform hydrostatic state of stress produces the same specific change of length in every direction. Whether the material is isotropic or anisotropic, there will be a change of volume. This change of volume can be perfectly elastic (which seems usually to be the case with materials which are homogeneous so long as we do not descend to molecular dimensions). It may also be partly or wholly of a permanent character.

A deviatoric state of stress produces a deformation in which the specific change of length has not the same value for every direction. Angles between directions fixed in the material will, in general, be altered. This deformation may be accompanied by a change of volume, although this is not an essential feature. Again the deformation may be either perfectly elastic, or partly or wholly of a permanent character.

Consequently a distinction can be made between elasticity of volume and elasticity of shape: the two need not occur together.

For the study of the phenomena of viscous and plastic deformation, the problem as to whether a material exhibits elasticity of shape is usually of greater importance than whether there is elasticity of volume. In the present Report the term "elastic deformation" will be used in the case where a deformation due to a deviatoric state of stress exhibits a wholly or partially elastic character. In cases where it is necessary to consider also the effect of a uniform hydrostatic stress, the magnitude of the hydrostatic pressure or tension can be introduced as a separate parameter, as is done with temperature.

10. Deformation. – In order to recognise the presence of a change of form, a comparison between two configurations is necessary. The deformation is defined by the difference in relative position of certain points or elements of the material in the state under observation, in relation to their relative positions in an "original state". Consequently the description of a deformation is of a different nature from that of a state of stress.

The total deformation is determined by the components of displacement, ξ, η, ζ of any material point as functions of the original coordinates x, y, and z, of that point. The coordinates of the point in the displaced state are then given by

$$x' = x + \xi, \qquad y' = y + \eta, \qquad z' = z + \zeta$$

Equation 3

(Greek letters ξ, η, ζ have been taken for the displacement in order to be able to use u, v, w as components of the velocity; compare section 15. The use of η is unavoidable in this connection although later on it will be used to indicate the coefficient of viscosity. It is expected that there will be no danger of mistaking one quantity for the other).

It will be understood that constant values of ξ, η, ζ throughout the whole specimen are of no importance for subject.

Hence constant amounts may be added or subtracted, so as to make ξ, η, ζ all equal to zero at some central point of the specimen. The "total deformation" is then defined relative to this point. We shall always assume that this is done.

A deformation is called "small" when the specific changes in length are small compared with unity, which requires the derivatives of ξ, η, ζ with respect to x, y, z, to be small compared with unity. Small deformations can be superposed independently of the order in which they are combined. This property makes possible a simple analysis of small deformations into elementary types.

Deformations of finite magnitude cannot be superposed in a simple way, since, except in special cases, the result is dependent on the order in which they are introduced.

11. A homogeneous or uniform deformation is obtained when ξ, η, ζ are linear functions of x, y, z (this definition remains valid when the displacements are not small). In such a deformation all pairs of points which are initially at equal distances from one another in parallel lines, remain equidistant and in parallel lines, the distance possibly being altered and the direction of the lines changes (W. Thompson and P. G. Tait, Natural Philosophy, Part I, art. 155).

The simplest types of homogeneous deformations are:

(a) the deformation produced by uniform hydrostatic stress;
(b) uniform linear extension or compression;
(c) uniform shear.

Case (a) is of less importance for the characterisation of rheological behaviour. – With regard to (b) it is to be observed that extension (or compression) in the direction of one coordinate can be accompanied by opposite deformations in the direction of the other coordinated. Hence the general expression for this case is given by the equations

$$\xi = ax, \qquad \eta = -by, \qquad \zeta = -cz$$

Equation 4

where a, b, c, are constants. By taking special measure it is possible to influence the values of b and c; e.g. the case c=0 can be realised by clamping the sides of the specimen in an appropriate way.

In the ordinary extension experiments carried out on a cylindrical rod, the constants b and c will be equal, so that the deformation is described by:

$$\xi = ax, \qquad \eta = -by, \qquad \zeta = -bz$$

Equation 5

In case (c), two sub-cases will be distinguished:

pure shear, defined by

$$\xi = bx, \qquad \eta = -by, \qquad \zeta = 0$$

Equation 6a

and simple shear, defined by

$$\xi = ay, \qquad \eta = 0, \qquad \zeta = 0$$

Equation 6b

Pure shear has been defined here with reference to coordinate axes parallel to the principal directions of extension and contraction, so that it appears as a special type of case (b). In pure shear there is no rotation of the elements of volume. In Equation 6ab for simple shear the coordinate system is defined so as to make the x-axis parallel to the direction of displacement. A simple shear is equivalent to a pure shear superposed on a rotation.

Unlimited continuation of simple shear is possible when the material can be brought into a state of laminar motion between two parallel walls of unlimited extent. In practice this will require an unlimited supply of material. Usually in the practical investigation of the behaviour of a material is subjected to shear, one has to be content with an approximation to the state of simple shear if a state of infinite laminar motion cannot be realised.

12. Other types of deformation of relatively simple character are:

(a) axial shear in concentric cylindrical layers, the elements of the material moving parallel to the common axis, provided the displacement u is a function of the distance r from the axis only (example: the laminar motion of a liquid in a cylindrical tube);
(b) tangential shear in concentric cylindrical layers, the elements of the material moving in circles lying in planes perpendicular to the axis of the cylinders and having their centers on the axis, the angular displacement being a function of the distance r from the axis only;
(c) torsion, in which case the elements of the material move likewise in circles in planes perpendicular to the common axis having their centers on the axis, the angular displacement being a linear function of the coordinate z measured along this axis

In all these cases, as in that of simple shear, the displacement is in a direction perpendicular to the coordinate on which depends the magnitude of the displacement, so that this coordinate itself does not change during the process. – In cases (b) and (c) an unlimited deformation can be realised with a finite quantity of material.

In none of these cases, however, is the deformation homogeneous as it is with simple shear

Somewhat more complicated instances of deformation are obtained by combining the rotational deformations (b) and (c) with axial extension or compression

13. The internal stresses appearing in a material subjected to a change of form, will depend on the differences in magnitude of the components of displacement ξ, η, ζ in points of the material situated at distances from each other comparable with the dimensions of the structural elements of the substance.

When the structural elements are of microscopic or of molecular size (so that they are small compared with the distances over which there is an appreciable change of ξ, η, ζ), the stresses will depend exclusively on the local deformation of the material, which is to be calculated from the first derivatives of ξ, η, ζ with respect to x, y, z. This is the case usually considered.

One can imagine, however, materials in which there exist structural elements (e.g. fibers) extending over finite lengths, comparable with the distances over which even the first derivatives of ξ, η, ζ change appreciably. The material can still be macroscopically homogeneous in the sense defined in section 1, provided the fibers are distributed sufficiently homogeneously throughout the material. In such cases complicated relations for the stresses will be found, in which finite differences of the components ξ, η, ζ may play a part.

In many textbooks the local deformation is termed the strain. Certain authors restrict the term strain to the recoverable part of the local deformation, on the basis of the assumption that it is this part in particular which is responsible for the appearance of elastic stress. The distinction between the recoverable and the non-recoverable parts of a deformation can only be made by means of special experiments. We therefore prefer not to introduce this distinction at the present time and will keep to the term “local deformation”.

14. In the case of small deformation, the local deformation is defined by the symmetrical tensor of the second rank;

$$\left|\left|\varepsilon\right|\right|$$

Equation 7

with components:

$$\varepsilon_{xx} = 2\frac{\partial \xi}{\partial x}, \qquad \varepsilon_{xy} = \varepsilon_{yx} = \frac{\partial \eta}{\partial x} + \frac{\partial \xi}{\partial y}, \qquad \text{etc.}$$

Equation 7a

These components can be divided into two groups:

$\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{zz}$ (or, for shortness, $\varepsilon_x, \varepsilon_y, \varepsilon_z$): extension

$$\varepsilon_{xy} = \varepsilon_{yx},\ \varepsilon_{xz} = \varepsilon_{zx},\ \varepsilon_{yz} = \varepsilon_{zy}\text{: shear}$$

Equation 8

Remark:- There is some difference of convention with reference to the notation of the components of the local deformation, as certain authors use half the values given here either for the symmetrical or for all components. The notation of Equation 8, which can be contracted into a single formula

$$\varepsilon_{ij} = \frac{\partial \xi_i}{\partial x_j} + \frac{\partial \xi_j}{\partial x_i}$$

is used for convenience and in analogy with hydrodynamical usage (compare Equation 11); it is not as yet to be taken as a final proposition to be prescribed for general use.

In the case of simple shear as defined by Equation 6ab the components of the local deformation become:

$$\varepsilon_{xx} = \varepsilon_{yy} = \varepsilon_{zz} = \varepsilon_{xz} = \varepsilon_{zx} = \varepsilon_{yz} = \varepsilon_{zy} = 0, \qquad \varepsilon_{xy} = \varepsilon_{yx} = a.$$

For shortness the letter γ will be used in this case to denote the component ε_{xy} so that

$$\gamma = \frac{\partial \xi}{\partial y} = a$$

Equation 9

15. When the magnitude of deformation changes with time, the rate of the change becomes important.

The total rate of deformation is defined by the time derivatives of the displacement components. These will be denoted by u, v, w (components of the velocity of displacement), so that

$$u = \frac{d\xi}{dt}, \qquad v = \frac{d\eta}{dt}, \qquad w = \frac{d\zeta}{dt}$$

Equation 10

Again (as in section 10) It will be evident that constant values of u, v, w throughout the whole specimen will be of no importance. Constant amounts may always be added to or subtracted from these quantities so that we may assume that u, v, w are made zero at some central point of the specimen.

The rate of deformation will affect the magnitude of the stresses in the material through the combined effect of progressive deformation and molecular re-adjustment. In an important class of substances an equilibrium between these two phenomena can be reached when the rate of deformation is independent of the time or does not change too quickly. There will then result a definite distribution of positions and velocities of the molecules (or atoms), differing form the

statistical distribution to be found in the normal state, so that we can say there exists in the material a definite state of internal "physical deformation". The physical deformation in this case is not given by the total deformation as specified by the values of ξ, η, ζ, which are increasing with time; but is determined by the difference between the actual statistical state of the molecular field and the normal state, which difference exhibits a stationary character.

Since this difference will increase both with the rate of the deformation and with the length of the time of re-adjustment of the substance, its effect on the stresses appearing in the material will be comparable to that of a deformation, determined by the product of these two quantities. In the case of large re-adjustment times more complicated formulae may be required, in which the time integral of the rate of deformation multiplied by some "memory" function may play a part. However, keeping to the case of short re-adjustment times, we may conclude that the physical deformation, apart from a factor having the dimensions of a time and dependent on the molecular structure of the material, will uniquely be dependent on the values of the velocity components u, v, w, considered as functions of the coordinates x, y, z, of the point where they are measured. It follows that in this case the "local physical deformation", apart from the same factor, will be determined by a "local rate of deformation", to be calculated by applying to the quantities u, v, w similar formulae as those given in 1514

Consequently the components of the local rate of deformation are given by

$$D_{xx} = 2\frac{\partial u}{\partial x}, \qquad D_{xy} = D_{yx} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}, \qquad \text{etc}$$

Equation 11

These expressions for a symmetrical matrix, which is a tensor of the second rank:

$$||D||$$

Equation 12

and we can distinguish between "rates of extension" and "rates of shear"

In the case of simple shear we have to consider a single component only:

$$\frac{\partial u}{\partial y} = \frac{d\gamma}{dt}$$

Equation 13

When the time of re-adjustment is large, these formulae must be replaced by more complicated ones

16. Relation between stress and deformation. – The general problem of the relation between stress and deformation is extremely complex. The complexity arises from the fact that a deformation may produce a change of structure which either remains permanently or more or less disappears in the course of time in consequence of re-adjustment phenomena. Hence, whether it is required to find the resulting deformation when the applied stress system has been

given as a function of coordinates and of the time, or to find the stresses arising in consequence of a prescribed course of deformation, the expression of the desired relation may require the introduction of integral formulae, in which "memory" functions may play a part. It is only in special cases that the general functions play a part. It is only in special cases that the general relation will simplify to expressions which can be handled with some ease.

It is not our object to attempt a more complete analysis and, for the development of an adequate nomenclature, it is far more important to restrict ourselves to cases where the geometrical relations are simple. For this purpose we will choose the simplest case of homogeneous deformation accompanied by a homogeneous field of stress: simple shear. In general we shall take the stress as given and will treat the deformation as the consequence of the application of the stress. Moreover, we shall restrict the time-dependence of the stress to a few simple types.

17. State of stress accompanying simple shear. – When a material is subjected to a simple shear as defined by Equation 6ab we can assume that there will always appear a shear stress p_{yx} in the direction of the x-axis, acting across planes perpendicular to the y-axis. Considerations of mechanical equilibrium show that simultaneously there appears a shear stress p_{xy} parallel to the y-axis, acting across planes perpendicular to the x-axis, of the same magnitude per unit area. For simplicity we denote the magnitude of these stresses by τ.

In all experimental arrangements in which homogeneous shear is realised, the shear stress across the planes perpendicular to the y-axis is balanced by external forces applied to the system in such a way that the magnitude of τ can be measured. The shear stress acting across planes perpendicular to the x-axis is taken up by restraints in such a way that it does not affect the measurements.

Simultaneously with the shear stresses p_{yx} and p_{xy} other components may appear. This may be the case with anisotropic materials, and also with originally isotropic materials as a result of non-linear relations between stress and deformation. In the same way it is possible that when a specimen is subjected to a shear stress, the resulting shear will be accompanied by a change of longitudinal or of lateral dimensions. To this class belongs the remarkable phenomenon described by K. Weissenberg and others, of a fluid in which a tensile stress along the lines of flow are of importance and their analysis will be useful for purposes of detailed classification, we shall not enter into this analysis, and in what follows we shall restrict the description of the results to the relation between τ and the magnitude and course of γ as a function of the time.

In the cases of deformation mentioned in section 12 sub (a) and (b), there is a shear stress acting across cylindrical surfaces, while in (c) there is a shear stress acting across planes perpendicular to the z-axis. In every element of volume these stresses are again accompanied by stresses of equal magnitude acting across planes perpendicular to the direction of motion, which are not considered in making measurements. The stresses, however, are not uniform throughout the material, but are functions of the distance from the axis.

It is sometimes possible theoretically to deduce the magnitude of the stresses in cases (a), (b) and (c) when the relation between deformation and stress in the case of simple shear is known over a sufficient range. Also inverse relations may be deduced.

Part B: CLASSIFICATION OF FORMS OF RHEOLOGICAL BEHAVIOUR ACCORDING TO RESULTS DEDUCED FROM EXPERIMENTS IN WHICH THE MATERIAL IS SUBJECTED TO A UNIFORM AND CONSTANT SHEAR STRESS

Chapter III – Classification based on the magnitude of the permanent deformation.

18. In this Chapter we shall take for our standard method of investigation, experiments in which the material is subjected to a homogeneous shear stress of constant magnitude τ during a specified length of time t. We assume the resulting deformation to be a simple shear. As mentioned in 17, the shear in certain cases may be accompanied by a change of longitudinal or lateral dimensions, but we shall leave this out of consideration.

The quantity to be considered will be γ, the magnitude of the shear as a function of time,

(a) during the period t while the sample is subjected to the constant stress of magnitude τ;
(b) after the stress has been taken away.

It must be noted that the time will enter into our considerations in two different ways; as the total duration of a completed experiment (the period t), and as a quantity which continually increases during the experiment. To avoid ambiguity we shall indicate the latter quantity by t'; so that during a single experiment t' increases from 0 to t.

The first point to be considered will be the relation between the permanent deformation (see below) and t. In sections 2424 and 27 the dependence of γ on t' will call for attention. Further, in section 32, we shall use t'' to denote the time elapsed after removal of the stress, so as to be able to investigate the behaviour of γ as a function of t''.

The application of the force must satisfy the conditions stated in section 4: the removal must satisfy similar conditions. It will be assumed that the whole process is carried out isothermally. Although theoretically adiabatic conditions might be equally well chosen, the fact that, with progressively increasing deformation, there may be a continuous production of heat leading to a corresponding rise in temperature, makes the physical interpretation of the results less simple in the latter case.

After taking away the force(s) acting on the specimen, the deformation can either remain unchanged, or may wholly or partly recover. That part of the deformation which is found to remain after a sufficiently extended lapse of time, will be called the non-recoverable or permanent deformation (permanent shear) γ_p. It is assumed that the observations can be extended over a sufficiently long interval in order that the final value of the shear is either actually reached

or can be calculated from the data. If the final value should be zero, there is complete recovery of the original form.

When it seems impossible to determine the value of γ_p accurately, the length of the period of observation after removal of force and the degree of accuracy obtained should be stated. When series of experiments are performed, these quantities should be kept constant or be varied in some regular way.

It is possible that the course of γ as a function of the time after removal of the stress can be influenced by the application of special stimuli, e.g. subjecting the material to rapidly alternating small stress or to shocks, increase of temperature, humidity, the action of solvents etc., by means of which the extent of the return to the original shape can sometimes be increased (compare section 29). It must be stated whether such means have been applied or not, so that the meaning of γ_p may be duly qualified.

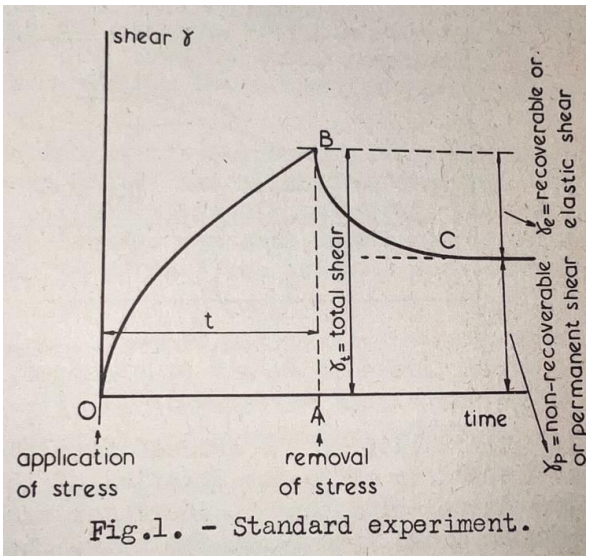


Figure 1 – Standard Experiment

19. The course of γ as a function of the time in an experiment of standard type, will be of general form, schematically indicated in fig. 1. Three quantities appearing in the diagram are of particular importance for our subject:

(1) The magnitude of the shear at the end of the period t, which is called the total shear

$$\gamma_t = \text{AB}$$

(2) the non-recoverable or permanent shear γ_p, as defined before, which is given by the asymptote to the curve BC;

(3) the recoverable or elastic shear γ_e, which is the difference between the total shear and the permanent shear: $\gamma_e = \gamma_t - \gamma_p$.

Actual detains of the diagram may vary wildly. In certain cases either γ_p or γ_e may be zero. With an ordinary liquid, the curve OB will become a straight line, while BC will be a horizontal line, so that γ_p is equal to AB= γ_t and $\gamma_e = 0$. In the case of an elastic solid, the curve OB will move very quickly upwards and then remain practically horizontal until the point B is reached; after the removal of the stress the curve will descend equally quickly and reach almost immediately to the basis of the diagram. In this case $\gamma_p \approx 0$ and $\gamma_e \approx \gamma_t$.

For the present we shall not consider the detailed form of the curves (to which further reference will be made in Chapter IV), and take from the diagram only the value γ_p of the permanent shear. This quantity is a function of the magnitude τ of the shear stress and of the length t of the period during which the stress has been acting. The temperature at which the experiment is carried out must be stated; if necessary also the pressure and humidity and

further the nature of the stimuli applied after removal of the stress, if there have been such stimuli.

We shall next consider a series of values of γ_p corresponding to different values of t, but all referring to the same value of τ and to the same external circumstances. Every individual experiment must performed on a fresh samples of the material. As we have assumed that the material is sufficiently homogeneous to make the experiments replicable, the curve γ as a function of the time t' during the period when the material is subjected to the stress, must all be segments of one single curve. This has been indicated in Figure 2 in which the curves, obtained from a series of standard experiments (all referring to the same value of τ), have been superposed.

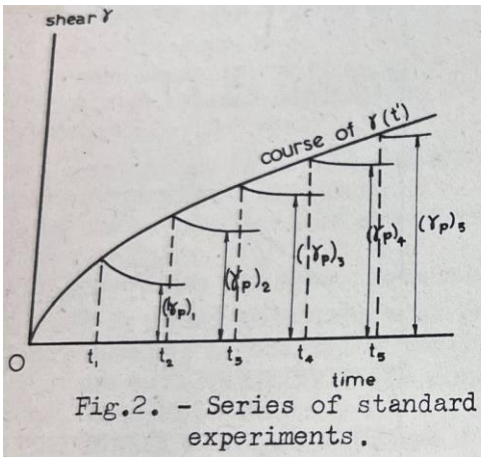


Fig.2. - Series of standard experiments.

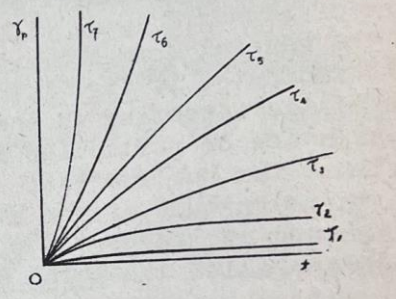


Fig.3. - Flow-diagram.

Figure 2 – Series of standard experiments.

Figure 3 – Flow-diagram

The values of γ_p deduced from these curves will now be considered as a function of t, the duration of the period of action of the stress. When it is found that γ_p increases with the duration t, we say that the substance exhibits flow. The curve obtained by representing γ_p as a function of t will be called "$\gamma_p - t$ curve" or "flow-time curve", or for shortness simply "flow curve". It illustrated the flow of the substance under the action of a definite value of τ.

Flow-curves evidently can be obtained for a series of values of τ. We thus are able to construct a set of flow-curves, constituting a "flow-diagram", a schematic example of which has been indicated in Figure 3. It must be observed that more complex shapes of the curves, including curves with inflections, are possible and that intersection of curves is not to be excluded on a priori grounds.

This set of curves, although not embodying all the rheological data apparent from the behaviour of the substance as it is observed in the experiments described, will be used as a basis for the classification of forms of rheological behaviour. The set refers to a single temperature, other external conditions also clearly defined. A more complete characterisation

of can be obtained when such sets are derived for a series of temperatures, but we shall not enter upon an investigation of the temperature influence, which forms a subject itself.

It should be noted that the term "flow-curve" as proposed here differs from the concept of "Fliesskurve", used by W. Phillippoff (see section 61).

20. First principle of classification, based on the character of the flow-curves for small values of τ. With a large class of substances it is observed that for a value of τ, either up to the breaking point, or below a certain threshold τ_f, the values of γ_p, even for large values of t, remain either zero or at least so small as to be almost unobservable. Our first distinction between types of rheological behaviour will be connected with this fact.

As a measure of the permanent deformation, we may compare it with the elastic deformation γ_e. Permanent deformations will be called "small", when they are small in comparison to the elastic deformation.

We shall use the terms perfectly elastic body, perfectly elastic behaviour, and perfectly elastic deformation if γ_p is strictly zero for a certain range of values of τ, so that the total deformation AB (Figure 1) appears to be fully recoverable. The subject of elasticity is further considered in Chapter IV.

When the permanent deformation is not rigorously zero, but does not exceed a few percent (say at most 5%) of the elastic deformation, we shall say that there is almost perfectly elastic behaviour. The term "imperfectly elastic behaviour" may be used when the permanent part of the deformation amounts to more than 5% of γ_e, but is still less than γ_e itself.

As the magnitude of the permanent deformation may increase with the duration of t of the experiment, a time-limit T must be fixed. By way of example we may that this limit T to be 3 hours, but this value may be replaced by a different one for particular classes of substances.

When the magnitude of γ_p becomes greater than γ_e, there is another form of behaviour, to be subdivided into two types, plastic behaviour and fluid behaviour. In order to define these types it is necessary to consider the domain of the values of τ in which permanent deformations larger than the elastic ones make their appearance.

Starting from the case in which perfectly, almost perfectly or imperfectly elastic behaviour during a period equal to the time-limit T is found with low values of τ, it is possible either that this behaviour occurs for all values of τ up to the breaking point for shear, or that it occurs in a limited domain of values of τ only, to be followed by another domain in which $\gamma_p > \gamma_e$. In the first case we say that there is elastic behaviour up to the breaking point: in the second case the substance is said to behave as a plastic substance. The limit between the two domains of τ in the second case represents the threshold value τ_f alluded to in the first paragraph of this section. It is often observed that the value of γ_p remains comparatively small for values of τ below τ_f and not to near τ_f. When τ approached τ_f there is a rapid increase of γ_p, so that for $\tau > \tau_f$ the value of γ_p rapidly becomes much greater than γ_e. In the flow-diagram the flow curves for $\tau < \tau_f$ and those for $\tau > \tau_f$ will then form two sets, distinct in type: those of the first set will

remain very near to the horizonal axis, whereas those of the second set will reach appreciably higher. In the view of this fact the threshold is called the yield-point and the stress τ_f at which it occurs is called the yield-stress or sometimes yield-value. It is convenient to keep the same term also for cases where the distinction is less sharp and can only be made by means of a precise comparison of γ_p and γ_e.

It must be observed that many authors define the yield-stress in a different way: instead of comparing γ_p with γ_e and taking as a limit $\gamma_p = \gamma_e$ they introduce a numerical limit for γ_p, e.g. 0.0001. The yield-stress is then defined as the stress for which γ_p reaches 0.0001. When speaking of the yield-stress, it is necessary therefore to state in which way the limit has been defined.

It will be evident that whatever definition of yield-stress is used, the time-limit T is of importance and must be stated likewise.

The yield-stress will, in general, be a function of the temperature, since all phenomena connection with permanent deformation are usually extremely temperature-sensitive.

Plastic behaviour, and likewise plastic deformations, are thus defined in connection with the behaviour of a substance over a range of values of the stress.

Substances which show perfectly or almost perfectly elastic behaviour in a certain domain of values of τ, are often called "solids". When a yield-stress is found, the term "plastic solid" is used (see also section 30).

21. In contrast to the case considered above, stands that where the values of γ_p, obtained with a period equal to T, even with the smallest applicable values of τ is larger that γ_e and thus is more that 50% of the total deformation. In this case, the curves of the flow-diagram do not show any characteristic difference in form for low values of τ and for higher values. Substances for which this is the case are said to behave as fluids (at the temperatures concerned).

In this case the resulting permanent deformations are called fluid deformations. It is not required that these deformations shall increase indefinitely with the duration of the experiment (a further distinction referring to this point will be developed in the next section). The decisive feature characterising fluid behaviour is the Absence of a yield-value for the stress. – Fluid deformations can still be accompanied by elastic deformations (section 31).

In the application of these definitions, the limit T is of much importance. This limit may be chosen differently in accordance with the phenomena which are considered to be important. For instance, glass behaves elastically up to the breaking point when our time-scale is of the order of some hours; whereas it behaves as a fluid when the time-scale is of the order of months or years. Other substances which behave as fluid in ordinary experiments, may show almost perfectly elastic behaviour when the time-scale is a small fraction of a second.

22. Second principle of classification, based on the form of these flow-curves for which the

permanent deformation is appreciable. We now turn to the consideration of the form of flow-curves. As stated in 19, whenever the permanent part γ_p of the shear increases with the period t during which the constant shearing stress τ had been acting, we say that the material exhibits flow. The velocity of flow or rate of flow is given by $d\gamma_p/dt$ and is thus determined by the rate of increase of the permanent part of the deformation only.

Since γ_p is a function of both the duration t of the experiment and the magnitude of the stress τ, it would be mathematically more correct to use the notation for the partial derivative $\partial\gamma_p/\partial t$, but for simplicity we have refrained from doing this, as no mistake seems likely.

According to the nature of the flow-curves the following distinctions between types of flow will be introduced.

1) Limited flow, leading to limited deformations, where the magnitude of γ_p does not exceed a finite limit, however long the shear stress is acting. Theoretically a further subdivision of these deformations might be possible, depending on whether $d\gamma_p/dt$ becomes rigorously zero after a finite lapse of time, or whether the limiting values of γ_p is approximated asymptotically after an infinite time
2) Unlimited flow, leading to unlimited deformations, where the magnitude of γ_p in the long run will exceed all finite values. If the velocity $d\gamma_p/dt$ decreases indefinitely, the flow will denoted as decelerating.
3) Uniform flow, leading to uniformly increasing deformations or liquid deformations, where the velocity of flow (either immediately or asymptotically) assumes a constant value, to be denoted by

$$D = \lim_{t=\infty} d\gamma_p/dt$$

Equation 14

4) Accelerating flow, where the velocity of flow increases with time. This may lead to some form of rupture of the interior structure of the substance. In particular a vertical tangles of the flow-curve may be considered as an indication of such happenings.

If desired, the term "complex flow" might be used in cases showing still greater complexity.

In certain cases it may be difficult to decide with certain whether a deformation must be considered as limited, or whether a very slow increase is still going on. In such case the term micro-flow may be applied. The rate may be decelerating, uniform, or perhaps accelerating.

It is to be observed that the appearance of limited (and also of decelerating) deformations is very often accompanied by hardening (compare section 37). On the other hand it should be noted that effects of "hardening" need not always lead to limited deformations, but may be accompanied by some form of much decelerated flow or by micro-flow.

For the term "creep", the reader is referred to sections 27, 32, and 60.

23. The nomenclature introduced here can equally well be applied whether a yield-value is present or not. In the first case, having regard to the principle that we are considering appreciable permanent deformations, we shall apply it in particular to the course presented by the plastic deformations, which appear when τ exceeds the yield-value. We may thus speak

(a) of limited, decelerating, uniform or accelerating fluid flow (when a yield-value is absent); or

(b) of limited, decelerating, uniform or accelerating plastic flow (when a yield-value is present). Uniform fluid flow will also be called liquid flow.

Substances exhibiting one of these types of flow in an important domain of values of τ will be said to behave:

1) where only limited deformations are obtained,
   (a) as a stiffening fluid (or semi-fluid); or
   (b) as a stiffening plastic substance;
2) in this second case
   (a) as a decelerating fluid; or
   (b) a decelerating plastic substance;
3) in the third case
   (a) as a true fluid (or liquid); or
   (b) as a uniformly flowing plastic substance;
4) in the fourth case
   (a) as an accelerating fluid; or
   (b) as an accelerating plastic substance.

If a yield-value is present and micro-flow is found to appear, we may likewise distinguish two regimes of micro-flow: micro-flow of the first kind (when τ is below the yield-value) and micro-flow of the second kind (when τ is above the yield-value). Micro-flow of the first kind consequently refers to the very slow increase of a permanent deformation, the magnitude of which during the time-limit T does not exceed the magnitude of the elastic deformation; whereas micro-flow of the second kind refers to the very slow increase of a permanent deformation of a magnitude exceeding the elastic deformation.

Remark. In so far as on the Earth all materials are subjected to the force of gravity, it may sometimes be difficult, in handling a particular sample, to prevent its coming under the action of accidental shear stresses. This may cause a certain difficulty in applying our definitions (which all start from experiments performed on "fresh" samples) in a strict way. It is usually assumed that these accidental shear stresses do no harm. However, in the case of a rather sensitive type of material, some uncertainty may arise concerning the proper interpretation of an experimental test. For instance, this might be the case with what has been defined in 1)(a) above as a stiffening fluid.

24. Third principle of classification, based on relations found when uniform flow is present

A third step can be taken, when uniform flow is found for a certain range of τ, so that flow-curves corresponding to values of τ within this range, approximate straight lines.

This case is thus characterised by the feature that a constant shear stress τ (in many cases only after a certain period of internal adaptation produces a constant local rate of deformation, the magnitude of which is given by the value of the flow velocity (rate of shear) D as defined in Equation 14.

This implies that an equilibrium is reached between deformation and re-adjustment, so that for every value of τ with in this domain considered, there is a definite and constant state of interior physical deformation

When this is the case it can generally be assumed that the magnitude of the elastic deformation γ_e (if this is present) becomes independent of the duration $\underline{t}$. The consequence will be that the curve fort the total deformation γ, considered as a function of $\underline{t'}$ (compare section 18), so long as the substance is subjected to the constant stress τ, will likewise approximate to a straight line. Moreover, the values of $d\gamma/dt'$ and $d\gamma_p/dt$ will become equal, if $\underline{t'}$ and $\underline{t}$ are large enough. Hence the value of the flow velocity $\underline{D}$, in the case of uniform flow, can equally be obtained form a $\gamma - t'$ curve.

To each value of τ there will correspond a certain value of $\underline{D}$. These values can be represented in a new diagram, in which they will form a segment of a curve. If for some (high) value of τ the uniform segment may end at some lower value of τ, if here uniform flow changes into accelerated flow, the segment ends. Likewise the segment may end at some lower value of τ, if here uniform flow should change into decelerating flow. This lower limit of liquid flow may, in certain cases, coincide with the yield point, but it is also possible that the actual yield-point is still lower. With ordinary liquids the lower limit for uniform flow is zero.

25. Substances showing uniform flow can consequently be characterised by a "$D - \tau$ curve", that is a curve giving the (ultimate) rate of flow as a function of shear stress. As the term "rate of flow-stress curve" is rather lengthy, we propose to call this curve "viscosity-curve" because it pictures that behaviour of the substance to which the concept of viscosity can be applied (section 26). The viscosity-curve is generally a segment, and refers, of course, to a definite temperature. Viscosity-curves can be obtained for a series of temperatures. Not only their course, but also their extent will in general depend on temperature.

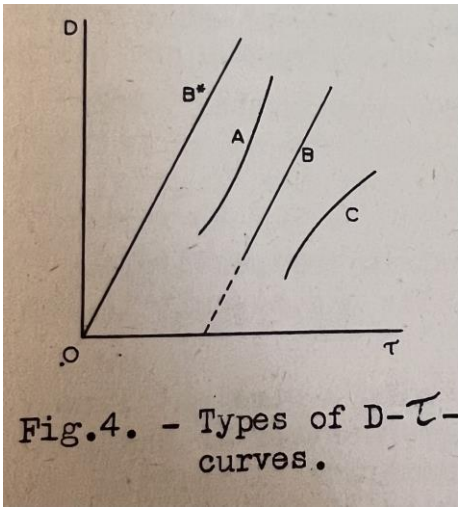


Fig.4. - Types of D-τ-curves.

Figure 4 – Types of $D - \tau$ curves

The following distinctions can often be introduced, according to the form of the viscosity curve (compare Figure 4):

(A) When the curve is concave upwards, we say that the substance exhibits flow with shear-rate thinning

(B) When the curve is a straight line, cutting the τ-axis at some positive value of τ (the occurrence of the intersection on the other side of the origin seems unlikely), the substance is said to exhibit Bingham flow.

(B*) When the curve is a straight line passing through the origin, the substance exhibits Newtonian flow. Substances which exhibit this type of flow for all values of τ are said to behave as Newtonian liquids.

(C) When the curve is concave downwards, the substance shows flow with shear-rate thickening.

More complicated forms may be possible, e.g. there can be curves with sigmoid shapes.

It must be observed that certain authors consider "thinning" (increase of $dD/d\tau$ with increase of τ) and "thickening" (decrease of $dD/d\tau$ with increase of τ) as effects due to the direct action of the shear stress, rather than effects determined by the magnitude of the rate of shear. This point may be of importance in those curious cases where $\underline{D}$ itself appears to assume a small er value with larger τ, which is an extreme form of thickening (see section 28 and Figure 5)

The arguments in favor of the one or the other point of view can not as yet be accepted as final, since too little is known about the changes of internal structure accompanying the flow of substances showing "thinning" or "thickening". The aim of the terminology proposed above is no more than to bear out that it is not the magnitude of the shear itself, but phenomena either due to, or responsible for, the rate of shear, which determine the form of the curve.

In cases (A) and (C) when it is desired to stress the absence or the presence of a yield-value, the adjective "liquid" or "uniform plastic" can be added to the word flow.

Bingham flow (case (B)) is a particular form of uniform plastic flow: Newtonian flow (case (B*)) is a particular case of liquid flow.

When the $D-\tau$ - curve is a segment, its extrapolation (either downwards or upwards) into domains of τ where uniform flow is not obtained, can only lead to misconceptions. According to the way in which the definitions of this Report have been framed, there can be no question of "non-reproducibility" or "hysteresis" in connection with $D-\tau$ – curve, because all our curves are deduced from experiments which are performed every time on a fresh sample, without repetition on the same sample.

We propose the use of terms like "pseudo-plastic flow", and "structural viscosity", "dilatant flow", etc. to be discouraged.

26. When a $D-\tau$ – curve exists, the ratio τ/D for any point of this curve is called the coefficient of viscosity:

$$\eta = \tau/D$$

Equation 15

The value of this ratio will refer in general to one particular value of τ (or one particular value of $\underline{D}$) only. Its c.g.s-unit is the Poise.

In the case of Newtonian flow ($D-\tau$ −curve being a straight line through the origin), the value of η becomes independent of τ and D, and described a simple physical property of the material exhibiting this type of flow. It will, of course, be a function of the temperature. We may thus speak of "normal viscosity" or "Newtonian viscosity" when η is independent of τ and D; and of "anomalous viscosity" in the general cases.

Some authors have proposed in the general case to call τ/D "average viscosity" and to introduce as well the derivative $d\tau/dD$ as "differential viscosity". It must be observed, however,

that the use of $d\tau/dD$ suggests that both τ and $\underline{D}$ should have additive properties. Model systems show that this cannot always be the case for τ and $\underline{D}$ simultaneously.

Following Bingham's proposal, the reciprocal value of the viscosity i.e. the ratio D/τ, is often called the fluidity. In the present Report it is preferred to use the term "fluidity" to define the qualitative property of admitting permanent deformations for the smallest possible values of the stress. (In connection with the application of the term "fluids" to all substances exhibiting this property, see section 21). "Fluidity" should thus bear the same relation to "plasticity" as "fluids" do to "plastic substances". It is therefore proposed not to use the term fluidity for the inverse of the viscosity, but to call this quantity simply: reciprocal viscosity (or, coefficient of reciprocal viscosity). The c.g.s.-unit of reciprocal viscosity has the generally accepted name of "Rhe" (reciprocal value of 1 Poise).

27. When only a single experiment is carried out in which γ is observed as a function of $\underline{t'}$ for a constant stress τ, and later (after removal of the stress) of $\underline{t''}$, the information available about the rheological behaviour of the substance is really very scanty. It cannot be considered as giving a sufficient basis for a description of this behaviour. Only one single curve of this type shown in Figure 1 is obtained. From this we can deduce the values of γ_p and γ_e for one single period of stress application, but no data are available concerning the actual dependence of γ_p of $\underline{t}$, that is concerning flow in the sense in which this concept has been defined in 22.

Nevertheless, in many cases one must be content with the determination of $\gamma - t'$ - curves, for constant values of τ, without being able to make a continuous distinction between recoverable and non-recoverable deformation. Moreover, in many practical problems it is the total deformation which really counts.

When it can be assumed that the recoverable (elastic) part of the deformation is of minor importance in comparison with the non-recoverable part, it may often be supposed that, when the experiment is continued over a period of sufficient length, the magnitude of the elastic part will attain a constant value. When this is the case we can write

$$\gamma_p(t) \approx \gamma(t') - K, \qquad \text{when } t = t'$$

where K is constant, so that,

$$\frac{d\gamma_p}{dt} \approx \frac{d\gamma}{dt'}$$

Equation 16

This means that the $\gamma - t'$ - curve will give approximate data on the flow properties of the material. The argument ahs already been used in 24, in the particular case where uniform flow is obtained. In that case there is good reason to suppose that a definite state of physical deformation is reached, depending on the rate of deformation. With non-uniform flow one cannot be fully certain about this point.

It is safe, therefore, to apply the terms "apparent flow" and "apparent flow-curve" to the $\gamma - t'$ - curve when the latter is used to describe flow properties. We can then distinguish

between "limited", "decelerating", "uniform" or "accelerating apparent flow", in analogy with the distinctions made in 23. Following the same trend of thought we shall use the term "apparent viscosity" for the ratio

$$\eta_{app} = \frac{\tau}{d\gamma/dt'}$$

Equation 17

This ratio will be a function of t' when the apparent flow-curve is not a straight line.

In experiments with so-called solid materials (e.g. metals) observation likewise often to the determination of $\gamma - t'$ - curves. A point of importance is the very slow increase of total deformation which may go on for long stretches of time and which is often denoted by "creep". Again in these experiments no distinction is made between the course of the recoverable and the non-recoverable parts of the total deformation, and there are certainly many cases where the recoverable part is appreciable. We should speak of "apparent micro-flow"

In practice the term "creep" is often applied in this case, but as this term is also used in a different sense (see sections 32 and 60), the qualification of "apparent creep" is recommended.

28. There are certain substances which show intersecting flow curves, in the way roughly indicated in Figure 5 E.g. a suspension of quartz particles of suitable concentration allows

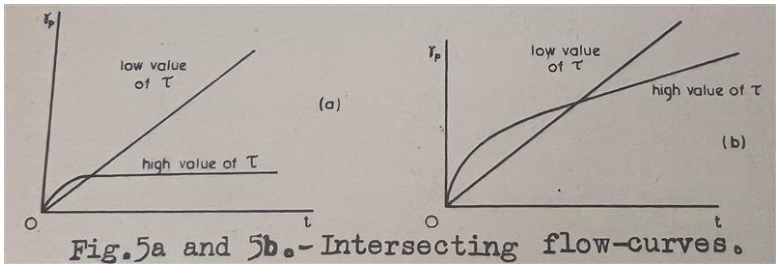


Figure 5a and 5b. – Intersecting flow curves

indefinitely increasing permanent deformations with very small stresses and then behaves as an ordinary fluid, or even as a liquid: where the application of large stress appears to produce a small limited deformation only, after which the substance seems to be solidified. (A considerable further increase of stress may possibly lead to rupture).

The appearance of increased resistance is probably connected with the rate of deformation, which may be expected to have a high value in the first instants after the application of a large shearing stress. This phenomenon has been much less extensively investigated that those described in the preceding sections. In most experiments carried out on this phenomenon a uniform distribution of stress throughout the sample and a homogeneous deformation have not been ensured; hence it is possible that inhomogeneity of the field of motion or accelerations play a part. These are circumstances which fall outside the range of our present considerations.

This property is sometimes called "dilatancy", on account of what is often a merely superficial likeness to a phenomenon described by Osborne Reynolds for wetted sand, in which the change from close packing to a packing with a greater pore space plays a part. In the case ofe the suspensions, however, there is frequently no overall change of volume, so that the term "dilatancy" is inappropriate. To avoid confusion a new term should be adopted for which we propose: shear-rate blockage, when the curve starting with the higher rate of shear becomes horizontal (Figure 5), meaning that movement stops completely.

There is also the possibility that γ_p continues to increase, the rate of increase, however, being smaller for higher values of τ, as exhibited by the curves sketched in Figure 5b. If in such a case the curves would approach to straight lines, so that uniform flow is obtained, we could construct a $D - \tau$ – curve or viscosity-curve, which would show a decrease of D with increase of τ. Such a case might be considered as an extreme form of shear rate thickening, in which the viscosity-curve turns completely downward, as was mentioned in Section 25.

Remark: The cases considered in sections 2020 –28 do not constitute an exhaustive classification of the forms of relationship between permanent deformation, stress and time during which the stress is acting: only the more common types have received attention.

Nevertheless, sets of flow-curves constructed for different temperatures and for each temperature referring to a complete series of values of τ, fully specify those basic aspects of rheological behaviour, which have been considered in this Chapter. These aspects, and the classification based on them, have been summarised in the Table accompanying this Report.

Chapter IV – Recovery of shape and effects of retardation.

29. Recovery of shape. – In sections 18 and 19 we started with the consideration of a curve, giving the deformation γ as a function of the time (t') for an experiment in which the specimen was subjected to a shear stress of constant magnitude during a period t, after which the stress was removed. Our attention was mainly directed to the magnitude of the non-recoverable or permanent deformation (permanent shear) γ_p, as found after a sufficient lapse of time following the removal of the stress. We have defined the recoverable or elastic deformation γ_e as the difference between the total deformation γ_t(=AB) and γ_p. We now turn to a consideration of some details of the recovery of shape or elastic recovery.

It will be evident that recovery of shape can occur in connection with all possible forms of curves γ as a function of t' (curves of the type as given in Figure 1). Hence elastic deformations can be obtained with both substances behaving like solids and with substances behaving like fluids or liquids.

Internal friction of a particular kind may sometimes prevent recovery to a greater or smaller degree, while nevertheless a more or less complete return to the original shape can be obtained by the application of a special stimuli, e.g. subjecting the material to rapidly alternating small stresses or to small shocks, increase of temperature, the action of solvents etc. Reference to this phenomenon had been made in the last paragraph of 18. In such a case we may say that there is a latent or blocked elastic deformation.

The recovery obtained on suitable stimulation is termed "stimulated recovery" or "redress" (mechanical redress, thermo-redress, etc. as the case may be): the recovery obtained without stimulation is called spontaneous recovery.

When the sum of the spontaneous and the stimulated recovery is such that the original deformation completely disappears, we may still speak of a perfectly elastic deformation, although with a latent part. It is also possible that, after removal of the blocking effect, a permanent deformation remains, in which case both the spontaneous and the stimulated recovery are imperfect.

30. When a substance shows perfectly elastic behaviour, we may assume that, to each value of τ within the range of this behaviour, there will correspond one single limiting value of the elastic deformation (which in this case is the same as the total deformation). Hence γ_e will be a function of τ in which the time does not enter. This function can be represented in a diagram as given in Figure 6. If, for a certain range of stress, γ_e and τ appear to be proportional, the elastic behaviour is said to be Hookean fort that range (in the diagram it has been assumed that this is the case until the point p_1 of the curve). The ratio τ/γ_e then represents the shear modulus of the elasticity:

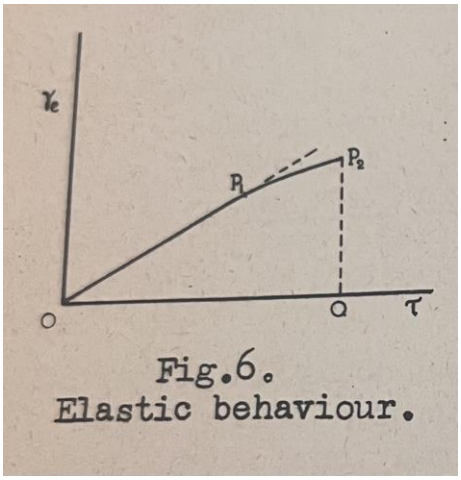


Fig.6.
Elastic behaviour.

Figure 6. Elastic behavior

$$G = \frac{\tau}{\gamma_e}$$

Equation 18

which will be a function of the temperature (although in general the elasticity modulus is less temperature sensitive than is the coefficient of viscosity).

The elastic behaviour is non-Hookean when the ratio τ/γ_e is not constant. In the diagram this is the case for the range p_1p_2. It has been assumed in Figure 6 that p_2 represents the limit of perfectly (or, as the case may be, of almost perfectly) elastic behaviour and that for higher values of τ permanent deformations appear which are not small in comparison with the elastic deformation. The distance OQ then represents the yield-stress.

As mentioned in 20, when there is a permanent deformation not exceeding a small fraction of the total deformation (even when this deformation slowly increases with time, so that the substance exhibits micro-flow), we speak of almost perfectly elastic behaviour. Data concerning the magnitude of the permanent deformation and concerning the periods of time allowed, must be given in order to make clear the meaning of the term "almost".

In the case where perfectly or almost perfectly elastic behaviour is found, a distinction can be made between ordinary elasticity (or, low elasticity) and high elasticity. In the first case the elastic deformations, even for shear stresses approaching the breaking limit do not exceed about one percent, while the modulus of elasticity is high of the order of $10^{11} - 10^{12}$ dyne/cm^2.

Materials showing these properties are called "solids" (perfect solids, almost perfect solids). When perfectly or almost perfectly elastic behaviour is found up to the breaking point, we speak of brittle solids. If a yield-value is reached before roaching the breaking point, we use the term plastic solid.

In the second case elastic deformations far exceeding 1%, sometimes reaching the order of several hundred percent, can be obtained. The modulus of elasticity may be of the order of 10^7 dyne/cm^2 or lower (e.g. flour dough $6 \cdot 10^4$ dyne/cm^2). Materials with high elasticity are not to be considered as solids in the strict sense.

The phenomena of high elasticity are due to mechanisms wholly different from those which appear in ordinary elasticity and are connected with the possibility of uncoiling of large chain molecules, which in the normal state are coiled up. Since this uncoiling is often a transition to a state of greater order, thermodynamic relations are involved. The most important of these is that for constant deformation, the stress will increase with temperature, if there is a decrease of entropy with increasing deformation at constant temperature. The relevant equations are usually developed for the case of extension. They fall outside the scope of the present Report.

31. In the case of a plastic material subjected to a stress exceeding its yield-value, and also with fluids and liquids, elastic effects can be found which are then combined with flow. E.g. with fluids or liquids elastic effects can make themselves felt in the form of a recoil, when the stress is suddenly reduced during flow. In consequence of the fact that flow increases progressively with time, whereas the elastic deformation will be of limited magnitude, and often relatively quickly assumes its final value, the time-scale of the experiments is of importance. With short durations elastic effects come to the fore: with long durations flow becomes proponderant (compare the last paragraph of 21).

The total deformation may be denoted as a "plastic-elastic", a "fluid-elastic" or a "liquid-elastic" deformation, as the case may be. Corresponding to these terms we can speak of fluid-elastic flow etc.

The term "flow-elasticity" will be used in particular in the case of fluids for which the elastic recoil (the recovery) is large. Materials exhibiting this phenomenon are called "flow-elastic". The elastic reaction in this case usually will have the character of high elasticity, with large extensions and low elastic moduli.

When elasticity is absent, we can either simply use the terms "plastic", "fluid", or "liquid deformation", as defined before in sections 20, 21 and 22 : or, if it is necessary to call attention to the absence of elastic effects, "plastic-elastic", "fluid-inelastic", or "liquid-inelastic deformation" respectively.

It is proposed not to use the terms "visco-elastic" and "visco-inelastic" in the general sense intended here, since viscosity can be defined only for uniformly increasing liquid deformations.

32. Retardation. - As has been indicated in Figure 1Figure 1 – Standard Experiment, recovery of shape often takes a certain

time. Restricting ourselves to the case where no stimuli are applied, we can then speak of "slow" or "retarded spontaneous recovery". The term "elastic after-effect" is also used.

When the recovery occurs almost instantaneously or when time effects are exclusively dependent upon inertia, the term "rapid spontaneous recovery" is used. We shall say that a substance shows "ideally elastic" behaviour, when there is perfect elasticity with (necessarily complete) rapid spontaneous recovery.

It is often observed that the recovery curve shows two parts, the first part corresponding to rapid recovery, and the second part (following upon the first part) corresponding to slow recovery.

Slow spontaneous recovery can be ascribed to phenomena caused by internal friction, depending on the value of $d\gamma/dt''$, where t'' is the time elapsed since the removal of the force. Where such internal friction is present, it is to be expected that it will make its influence felt whenever γ changes with time. Hence if again we use t' to denote the time continuously increasing from 0 to t during the period when the stress τ is acting, there must be frictional effects depending on $d\gamma/dt'$, which will affect the course of γ with time during this period. In particular this effect will retard the establishment of the elastic part of the deformation. In such cases we may speak of a "retarded elastic deformation". Sometimes the term "elastic fore-effect" is used to describe this phenomenon, although it is not a very satisfactory name.

Certain authors use the term "primary creep" to denote the slow establishment of an elastic deformation, and "creep recovery" to denote the slow elastic recovery, in cases where the retardation effects are large (compare section 60).

In many cases there exists a relation between the course of $\gamma(t')$ immediately after the application of the stress and that of $\gamma(t'')$ immediately after its removal. Theories have been worked out to describe such relations and model systems illustrating them have been devised. E.g. we may refer to "First Report on Viscosity and Plasticity", Acad. Sciences Amsterdam 1939, Ch. I, and to H. Leaderman, "Elastic and Creep Properties of Filamentous Materials", Washington, D.C., 1944, Part I. These theories, however, assume linear relations and consequently cannot embrace all possible cases.

33. Further remarks on frictional effects. - The frictional effects which we have encountered are partly:

a) dynamic friction, depending on the magnitude of $d\gamma/dt'$, or $d\gamma/dt''$, and disappearing when this magnitude is zero; and partly
b) static friction, remaining active when the rate of deformation is zero.

Static friction produces the blocking effects mentioned in section 29, which can be recognised as such if it is possible to remove these effects by means of suitable stimuli. Blocking effects will play a part not only in preventing spontaneous recovery, but can also be affected during the period of action of the force, with the result that in such a case the deformation will not always be univocally determined by the magnitude of the stress. The application of stimuli (e.g. the application of small alternating stresses or of small shocks) will then be necessary during the

period in which the force is being built up to its final value, in order to overcome the frictional effects and to bring the material into a state in which the elastic reactions are in equilibrium with the shear stress.

Dynamic frictional effects are dependent on the rate of deformation and disappear when the deformation retains a constant magnitude. If unaccompanied by static friction, they do not influence the ultimate magnitude of the deformation produced by a given stress, nor the degree of recovery after unloading. They influence only the time in which the deformation is produced or recovered.

The distinction between static frictional effects and frictional effects depending on the rate of shear, need not be a consequence of intrinsic differences in physical structure. In a number of theories concerning viscosity, the underlying idea is that shear stresses make their appearance in consequence of static effects connected with forces acting between the molecules. It is supposed that, in consequence of the heat motion, there are produced molecular displacements of such a nature that these stresses partly disappear, and that a shearing motion of certain layers of molecules becomes possible. In this way a relation between shear stress and rate of shear is obtained, in which the intensity of heat motion plays a decisive part.

It might be assumed that blocking effects are a primary feature of many classes of materials, and that in the case of sufficiently strong heat motion the blocking is overcome through internal re-adjustment and does not appear as such in the experiment. On the other hand, when the heat motion is insufficient, the blocking becomes noticeable; and increase of temperature, small shocks or vibrations, or chemical reactions may be required to overcome its influence.

In connection with the possibility of removing blocking effects by internal heat motion, it will be seen that we may not always assume the re-adjustment to take place so quickly that the remaining stress will be determined exclusively by the instantaneous rate of deformation. It is possible that stresses in the material will depend upon the difference between the present position of the molecules and their position some finite interval of time ago. It is to be expected that in such cases there will be a non-linear relation between stress and rate of deformation, while further complications may arise when the rate of deformation is variable. Reference to such phenomena has been made in section 15.

34. It may often be impossible to determine whether observed phenomena of retardation must be ascribed to dynamic friction dependent on the rate of increase of the deformation, or whether it is better to say that the deformation can increase in consequence of internal re-adjustment.

Two typical cases can be illustrated with the aid of differential equations. An equation of the type

$$\tau = G\gamma + \eta \frac{d\gamma}{dt}$$

Equation 19

may be used when we have elastic deformations, accompanied by internal friction causing retardation. In this case, the application of a constant stress τ at the instant $t = 0$ gives rise to the appearance of a deformation γ, determined by the formula:

$$\gamma = \frac{\tau}{G}\left(1 - \exp\left(-\frac{Gt}{\eta}\right)\right)$$

Equation 20

which for $t \to \infty$ approaches to the finite value τ/G . The ratio η/G in this case is called the retardation-time.

In other cases an equation of the type

$$\frac{\partial \tau}{\partial t} = G\frac{\partial \gamma}{\partial t} - \frac{\tau}{\lambda}$$

Equation 21

may be appropriate. The solution for the sudden application of a constant stress is:

$$\gamma = \frac{\tau}{G}\left(1 + \frac{t}{\lambda}\right)$$

Equation 22

leading to an indefinite increase of the deformation. Equation 21 can also be used to describe the simplest case of relaxation of stress, when γ is kept at a fixed value; the relevant solution will be given in section 46. The parameter λ is called the relaxation-time.

We may also have to do with cases where combinations of Equation 19 and Equation 21 would seem to be required, so that a number of both retardation- and relaxation-times would be introduced. Certain generalised types of equations in which an integration occurs over a whole spectrum of retardation- or relaxation-times are of great importance in the description of the behaviour of high polymers under a periodically varying stress (compare section 48).

Chapter V – Application of repeated experiments performed on a single specimen.

35. As stated in section 6 repeated tests performed on the same sample of material can be used to investigate whether the rheological properties of the material have or have not been changed by subjecting it to deformation. In most cases evidence can be obtained in the following way. Observe, first, the course of γ as a function of time (t') under the action of a constant stress τ, during a period t. After this, the stress is removed and the specimen is then left to itself for a specified interval of time, after which a second experiment is performed with the same value of τ. If an identical $\gamma - t'$- curve is obtained in the second experiment, so that we can say that the experiment is reproducible, this will usually mean that the rheological properties have not changed. If, on the other hand, the new $\gamma - t'$- curve differs from the original one there has been a change of the physical state of the material. The duration of the interval between the two experiments is of importance. In many cases it is found that with increase of this interval there appears some recovery of the original physical state: there may even be a complete recovery.

Recovery of the physical state is sometimes called restoration.

Recovery of the original physical state can also be stimulated. Consequently we can distinguish between spontaneous restoration and stimulated restoration.

Stimulated restoration can be produced, for example by subjecting the specimen to a higher temperature. However, such a process merges into a reconditioning of the material, and thus falls rather outside our subject (e.g. in metals recrystallisation can set in).

36. Amongst the most frequent types of changes of physical state, which become apparent from a repeated test, we can distinguish "hardening" and "softening", according to whether the repeated application of the same stress produces a smaller or a larger deformation than the original one.

In certain cases it is possible to find out whether the hardening or softening can be ascribed

(a) mainly to the magnitude of the total deformation to which the material has been subjected in the first experiment; or
(b) to an effect of the rate of deformation.

Special experiments can sometimes help in deciding between these possibilities. In such cases To can make a distinction between: "work hardening", "work softening"(as general terms), or more specifically: "shear hardening", "shear softening", on the one hand; and "shear-rate hardening", "shear-rate softening", on the other hand.

(It) is further possible that the change of physical state appears only when either the deformation or the rate of deformation exceeds some critical magnitude. In so far as this can be considered as a consequence of a stress exceeding a certain threshold value, we may speak of a yield-value for internal changes.

This new yield-value need not be the same yield-value as the one defined in section 20 in connection with permanent deformations. E.g. ordinary liquids like water have a yield-value of practically zero, when deformation is considered, and an almost infinite yield-value, when we consider internal changes. On the other hand, in metals the appearance of a permanent change of form is usually accompanied by a change of rheological properties, so that here the two yield-values are identical.

In all cases mentioned there is a further possible distinction, according to whether a spontaneous return to the original state occurs or not. If there is a spontaneous return, it can be complete or incomplete; also its time-scale is of importance. We can thus speak for example of reversible (complete or incomplete), or non-reversible work-hardening, etc.

37. In the preceding considerations we have met with various phenomena, which reveal an increase of resistance to deformation (or increase in "consistency" as it is often called) in materials:

(1) decelerating flow (section 22);
(2) hardening (section 36);
(3) thickening (section 25).

Decelerating flow is present when the flow-curve ($\gamma_p - t$ - curve), defined in section 19, is concave towards the axis for t (the horizontal axis in Figure 2), meaning that $d\gamma_p/dt$ decreases with increasing values t. According to the rules set forward, every point of the flow-curve should be obtained from a separate experiment, each experiment performed on a fresh sample. It has been indicated in section 27, that in those cases where it is impossible to execute a complete series of experiments, some information can be obtained from an "apparent flow-curve", giving γ as a function of t', as observed in one single experiment, although such information can never be fully decisive.

When decelerating flow is found, either from an exact flow-curve deduced from a complete series of experiments, or from an "apparent flow-curve", the decrease of the rate of flow is an indication of an internal change going on in the material. It is therefore to be expected that at the end of the experiment the material will be in a state differing from that at the beginning. Hence we must expect that when the experiment is repeated on the same sample, the new curve for γ as a function of t' will not be identical with the original one, but from the start will bear evidence of increased resistance to flow. This means that as a general rule, we may expect that decelerating flow will be accompanied by hardening.

What has been observed here with reference to decelerating flow naturally includes the case of limited flow.

Thickening, on the other hand, or more precisely shear-rate thickening, as it has been called in 25, refers to a property of the $D - \tau$ −curve or viscosity-curve, which can be determined only if there is uniform flow for a certain range of values of τ. The fact that shear-rate thickening is observed, does not in itself give any indication concerning a more or less permanent change of properties. The thickening can be present only so long as the rate of deformation has a sufficiently high value, and it can very well disappear immediately and completely, when the rate of deformation falls to zero. Hence with a substance exhibiting shear-rate thickening, it is still possible, that a repeated experiment will show exactly the same behaviour as the first experiment. When this is the case, there is no hardening effect.

A hardening effect may be obtained, however, with certain substances, and it may be found that the amount of hardening depends on the magnitude of the rate of shear to which the material had been subjected. In that case we should speak of shear-rate hardening, accompanying the shear-rate thickening.

38. Work hardening in plastic solids. - Many plastic solids exhibit an increased yield-value for deformation, when they are subjected to a second deformation process, after having suffered a permanent deformation. Permanent deformation in this case appears to be almost invariably connected with a change of rheological properties, so that we need not distinguish between two types of yield-value.

The phenomenon referred to is called "work hardening". In many cases it is practically non-reversible and can then be described in the following way:

(1) If a specimen is subjected in the first test to a stress below a certain threshold, there is perfectly (or almost perfectly) elastic behaviour. Permanent deformations are either completely absent, or so much smaller than the elastic deformations that they are hardly detectable. Repeated experiments performed upon a single specimen (if not repeated millions of times and at a high rate) do not alter the properties of the material, so that the deformations appear to be completely reversible and reproducible.

(2) with a stress exceeding a certain threshold value, permanent deformations appear comparable in magnitude with or greater than the elastic deformations. In many cases the threshold value is clearly marked: if not, it must be defined by fixing numerically a lower limit for the permanent deformations to be counted as such. The threshold value thus defined is the initial yield-value.

(3) Consider a case in which the specimen is subjected for the first time to a stress exceeding the initial yield-value by a finite amount, Suppose that the magnitude of the total deformation assumes a finite value, so that we have to do with a "limited deformation" as defined in section 22. On removing the external force producing the stress it is usually found that there is still a part of the deformation (the elastic part) which disappears; but there remains the permanent part, which is as a rule the larger of the two.

(4) On re-application of the external force it may now be found that the threshold value which must be exceeded in order to produce a new permanent deformation, is higher than the initial yield-value. There is thus a change of internal properties, which makes itself felt by the appearance of an increased yield-value. This new yield-value may reach almost the value of the maximum stress to which the specimen had been subjected in the previous experiment. Stresses remaining below this now yield-value give rise to elastic deformation only, and consequently can be repeatedly applied without alteration of the properties of the specimen. This phenomenon is the characteristic feature of work hardening.

(5) When the stress applied in the second experiment exceeds the new yield-value by a finite amount, a new limited permanent deformation is found to appear. When the force is again taken away, and a third experiment is then performed, this experiment may reveal a still higher yield-value, which may now reach the value of the maximum stress applied in the second experiment.

The stepwise increase of the yield-value may continue to a greater or lesser extent, until rupture sets in.

It is possible to distinguish in this way:
an initial yield-value, hardening yield-values and a limiting yield-value or "yield-limit" after which rupture appears.

39. The following observations are of importance in connection with the preceding description:

a) It will be seen that use has been made of a whole series of experiments in order to establish all the features enumerated.

b) The fact that in metals and in many other plastic solids there is usually no restoration, or only a very slight spontaneous restoration of the original physical state, when the interval between consecutive experiments is not too long, saved us from complications which might otherwise

arise. Periods of the order of hours or even days are allowable. Moreover the elastic part of the total deformation in metals usually disappears quickly after removal of the stress, without appreciable retardation; while after each application of the stress the deformation quickly reaches its final value.

c) As we have assumed that the permanent deformation produced by a stress exceeding the yield-value is limited (which is commonly the case with metals, where the small effects of micro-flow can ordinarily be neglected), it follows that to every value of the shear stress there corresponds a definite permanent deformation.

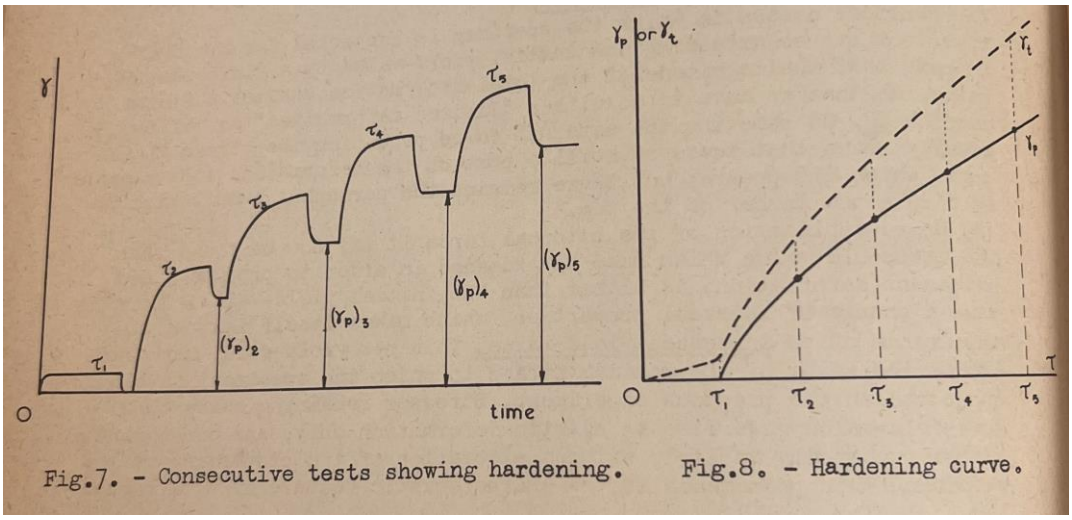


Fig.7. - Consecutive tests showing hardening.

Fig.8. - Hardening curve.

Figure 7 - Consecutive tests showing hardening.

Figure 8 – Hardening curve.

Figure 7 attempts to give a schematic picture of a series of consecutive tests with regularly increasing values of the stress and short intervals between the tests. The descending branches of the curve refer to the intervals when the force is removed. It has been assumed that the elastic deformation very quickly disappears every time the force is taken away.

The values of the permanent deformation γ_p obtained from such a series of tests have been represented in a separate diagram as a function of τ (Figure 8). The points have been connected by a curve, to be called the "real hardening-curve" (fully drawn line), which will be located more completely when the increase of τ is made in smaller steps. In the same diagram a broken line has been inserted, which represents the total deformation (sum of γ_p, and the elastic deformation γ_e).

The shape of the curves in the diagram is arbitrary. It is possible that γ_p at first increases very rapidly with τ, and afterwards more slowly, but other cases may be met as well. The curves are often drawn with the axes interchanged.

40. Experiments to reveal work hardening in metals, are often made in a different manner. Instead of repeatedly unloading the specimen in order accurately to distinguish between elastic and permanent deformation, the stress applied to the material is not taken away, but is gradually increased. Since every constant value of the stress produces a limited permanent deformation, it is to be expected that a slowly rising stress will produce a continually increasing deformation, such that at every instant the permanent deformation has nearly the value it would have if the stress were thereafter kept constant.

When the permanent deformations appearing in these experiments largely exceed the elastic deformations, no great error is made in considering the course of the total deformation as a convenient picture of the permanent deformation. The phenomenon of work hardening is then revealed:

(a) by the fact that a constant stress produces a finite total deformation, and that a larger total deformation can be obtained only by the application of a larger stress; and
(b) by the circumstance that, if at a certain stage of the process the external force is taken away, its renewed application reveals the increase of the yield-value above the initial yield-value.

It is to be expected that the course of the curve obtained in such an experiment will to a certain extent depend upon the rate of increase of the applied stress. Exact investigation has indeed revealed in some cases that a higher rate of increase requires a greater stress for the same magnitude of the total deformation. Hence, in general, different apparent hardening-curves will be obtained with different rates of increase. Moreover, in experimental work with existing testing machinery there are often influences due to peculiarities of the apparatus.

In order to obtain a well defined apparent hardening-curve it must be ensured that either the stress, or the deformation increases at a constant rate, which rate must be accurately specified.

A comparison of apparent hardening-curves obtained at different speeds can give data concerning phenomena of internal friction induced by the deformation process.

If the material shows micro-flow, its magnitude will influence the form of the apparent hardening-curve.

41. Instead of hardening, many substances exhibit a lower yield-value upon repeated loading. In this case we have softening.

Sometimes the yield-value may return to its original magnitude if the substance is left to itself after the experiment.

Many colloidal suspensions exhibit a change of rheological properties when they have been subjected to a deformation process. A common form is that the resistance to deformation decreases. Often it appears that the original rheological properties (and thus presumably also the original structure) are restored when the material is left to itself. The phenomenon has in this case been called thixotropy. It is proposed to expand this term so as to embrace all cases of

softening followed by a time-dependent return to the harder state: it is not necessary that there is a gel-sol transformation.

It is difficult to develop methods for the precise measurement of quantities which can be used as numerical data for describing thixotropy. The first point to be decided is whether a yield-value can be defied or not. In the second place we must consider the $\gamma - t' -$ cuirve for a value of τ sufficient to produce a permanent deformation. The phenomenon of thixotropy will reveal itself both.

(a) by the form of the curve, which will be concave upwards (accelerated flow), and will probably finally approximate to a straight line, representing uniform with a low value of viscosity; and
(b) by the fact that if no experiment is interrupted, there will be practically no elastic recoil, while if after a very short time interval the same stress is applied again, flow will appear with, from the start, practically the same high value of the velocity as was obtained at the end of the first experiment; and
(c) by the fact that if (after inturuption) the same stress is applied again after a long interval of time, the initial value of $d\gamma/dt'$ and the form of the $\gamma - t'$- curve will lie between that obtained on immediate reapplication of the stress (as in (b)) and that found in the first experiment (described under (a)).

The difference between the $\gamma - t'$- curve found in the first experiment and that obtained in the repeated experiment can be taken as a rough measure of thixotropy. A schematic picture has been given in Figure 9Figure 9 – Experiments on thixotropy. Curve (a) refers to the first experiment, performed with the substance in its original state (it exhibits the accelerated deformation). Curve (b) is obtained in a repeated experiment, with the same value of the shear stress τ as in the first one and with a very short interval of time between the two experiments. Curves (c) and (d) refer to experiments repeated after larger intervals of time.

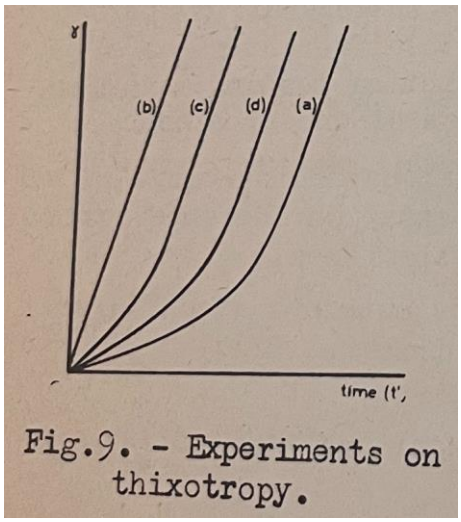


Fig.9. - Experiments on thixotropy.

Figure 9 – Experiments on thixotropy

The difference between the curves for a given interval of time between the experiments will usually be a function of the shear stress τ. Hence a more complete picture will be obtained by carrying out similar experiments for a series of values of τ.

It is also possible to start from the other side, first subjecting the substance to vigorous stirring, after which the $\gamma - t'$- curves are determined for a chosen value of τ with intervals of increasing magnitude between the stirring and the shearing experiment.

The time-scale characterising the return to the original condition is of great importance and can be used to define the “hysteresis” evident in the thixotropic behaviour.

It must be kept in mind, nevertheless, that the so-called "stirred" state of the thixotropic substance is never an equilibrium state and is always subject to a gradual change. It is only in those cases where the rate of restoration after thixotropic breakdown is very slow, that it may be possible to determine approximate viscosity-curves, for various intervals after complete stirring. These curves will then exhibit a gradual increase of the viscosity with increasing duration of the interval.

In some practical methods of investigation, the substance is first subjected to a gradually increasing shear stress, which is afterwards gradually reduced, and a curve is given for dy/dt' as a function of the stress. It is then found that the curve obtained with decreasing differs from that observed with increasing values of τ. On the whole, the former must be displaced in the direction of smaller τ (i.e., of higher values of dy/dt'). Such measurements, however, cannot give exact data, as the time-effects involved in the phenomenon enter in too complicated a way. Formulae have been developed for some special cases, but these formulae depend on particular assumptions and it is certain that they are inapplicable to all imaginable cases.

42. We have extended the meaning of the term thixotropy beyond that which is given to it by certain authors. From the rheological point of view it is nevertheless possible to distinguish between the following cases:

a) there is no yield-value in the original state, and likewise none in the "stirred" state;
b) there is a yield-value in the original state, which has disappeared in the stirred state (Freundlich's thixotropy);
c) there is a yield-value in the original state, and a lower one in the stirred state.

We propose to speak of thixotropic fluid behaviour in case a); of thixotropic plastic-fluid behaviour in case b); and of thixotropic plastic behaviour in case c). The names for substances behaving in one of these ways would be (a) thixotropic fluid substance; (b) thixotropic plastic-fluid substance; (c) thixotropic plastic substance.

It would seem that case b) represents Pryce-Jones' thixotropy, while c) apparently corresponds to what is called "False-body" (see section 52 under "Class D").

The effect of "stirring" on the rheological properties of a material is not always a decrease of resistance to flow (decrease of consistency). There are cases where stirring brings about increased consistency, that is, a form of hardening. Further with some thixotropic substances it has been found that the return to their original firmer state can be accelerated-by mechanical stimulation of certain kinds. This property is called “rheopexy”.

Chapter VI – Other types of experiment in which the material is subjected to shear

43. In the considerations of the preceding Chapters the magnitude of the stress was taken as an independent variable (the other one being time), while the deformation was the dependent variable.

In a number of experiments it is possible to proceed in a different way and to treat the deformation as an independent variable, on which the stress will be dependent.

Of special importance are the cases in which either the magnitude of the deformation or its rate of increase are kept constant during a certain period of time. The application of the latter method must in general be restricted to substances allowing of unlimited deformations, at least when there may be the possibility of indefinitely extending the time during which the rate remains constant.

It must be expected, of course, that there will exist certain relations between the behaviour exhibited by a substance in an experiment of the type mentioned here, and that where the stress is kept constant. However, it is impossible to give a mathematical formulation of these relations for all cases. Mathematical relations can be deduced when the behaviour of the substance can be described by means of linear differential or integral equations. But in the general case, the relation between deformation and stress will be non-linear, and the increase of the deformation for given increments either of time or of stress or both, may depend on the state of the substance at the instant considered. Further, this state may bear the marks of its whole past history, perhaps more or less effaced by complicated phenomena of re-adjustment. It then seems hopeless to predict the behaviour for, say, constant rate of deformation, when that under constant stress is known.

44. Experiments in which the deformation is kept constant. In order that the magnitude of the deformation may be kept constant during a certain period, a preparatory stage must have occurred in order to bring the deformation to the desired value. During this preparatory stage the applied stress will have passed through a range of values. When it is desired to reduce the length of the preparatory stage, high values of the stress may appear necessary in the first instants, so that we should then come into conflict with the requirement of section 4, that the stress shall increase from zero to its final value a monotonic way. To avoid this difficulty, it is better to suppose that, in the preparatory period, a constant value is given to the stress (having regard to the conditions mentioned in the fourth paragraph of section 4), so that the course of the deformation as a function of the time during this period will correspond to that depicted by the curve OB in Figure 1Figure 1 – Standard Experiment. From the instant A onward, the deformation having reached the value u, instead of taking away the stress as was supposed in Figure 1Figure 1 – Standard Experiment, we then keep the deformation constant (equal to AB), and investigate the behaviour of the stress.

It is possible that the stress remains constant, or that it decreases or increases. An increase of stress would point to structural changes occurring within the material. As such cases need special investigation and as it is difficult to foretell what may come of this, it is better to omit them here.

We may then state that, if sufficient time is allowed to elapse, the stress will finally assume a constant value, which may be the same as that applied during the preparatory stage in order to reach the deformation AB, or may be lower, and alternatively may be zero. It is to be expected that there will be some relation between the final value of the stress for constant

deformation, and the limiting value of the deformation after removal of the stress which was indicated in Figure 1Figure 1 – Standard Experiment by γ_p.

When the limiting value γ_p of the deformation after removal of stress is zero (perfectly elastic behaviour), the stress necessary to keep the deformation constant will be nearly the same as that applied in the preparatory period. There may be a difference when the increase of the deformation with time in the preparatory period would have produced internal dynamic friction (compare section 33). If, in this case, just before the instant A of Figure 1Figure 1 – Standard Experiment, the value of $d\gamma/dt'$ were not zero and part of the stress applied during the preparatory period were used to overcome such friction, the fixing of the deformation at a constant value would then eliminate this part of the stress, and the remaining stress would be below that applied in the preparatory period. The difference will disappear when the preparatory period is so long that $d\gamma/dt'$ has already become zero or when dynamic friction is practically absent. Rapid spontaneous recovery on removal of the stress would indicate the latter.

On the other hand, when the limiting value γ_p of the deformation after removal of the stress remains equal to AB, so that there is no recovery at all, we can be certain that, when the deformation is kept constant, the stress will fall immediately to zero.

Further, we might expect that whenever a substance undergoes indefinitely increasing permanent deformations for the smallest values of the stress to be applied (so that there is no yield-stress), the value of the stress for a constant deformation will always decrease to zero in the long run.

Inversely, when the stress for constant deformation does not, in the long run, decrease to zero, this indicates that the deformed state of the material is accompanied by internal stresses which do not disappear through internal re-adjustment. Hence we may conclude that a certain amount of recovery will appear when the external force is removed; and further that the substance, when subjected to a constant stress will undergo deformations of limited magnitude only, part of which is elastic.

45. The functional relations presented by the magnitude of the stress which is permanent at constant deformation, and the final deformation for constant stress (applied during a sufficiently extended period), will show close analogies. However, with substances showing effects of work hardening, it is possible that differences will make their appearance because the history through which the final state has been reached will not be identical for the two types of experiment, For instance, in a constant-deformation experiment the stress during the preparatory period may have been larger than the final stress.

The construction and investigation of various models (already alluded to in section 32) consisting of springs and dashpots, arranged partly in parallel, partly in series, can illustrate relations between the types of experiments which we are comparing; and a mathematical formulation can be given when it is assumed that all elements out of which these models are constructed, are governed by linear laws.

It is also possible to introduce effects of hardening into these models. However, when this is done and when, moreover, it is attempted to take account of non-linear relations, it is soon found that the number of possibilities becomes so large, that a considerable element of arbitrariness must be accepted, while at the same time mathematical treatment becomes extremely difficult.

46. Stress relaxation. - When, after fixing the magnitude of the deformation, there is found to be a more or less gradual decrease of the stress to some limiting value (in some cases zero), this phenomenon is called relaxation, or more precisely: stress relaxation. It is a particular form of the re-adjustment phenomena mentioned in section 2. It can often be described as a consequence of a gradual decrease of the recoverable or elastic part of the deformation with corresponding increase of the non-recoverable or permanent part.

In the simplest case the relation between the stress and the deformation is described by Equation 21 of section 34 The solution for constant γ is

$$\tau = G\gamma \exp\left(-\frac{t}{\lambda}\right)$$

Equation 23

indicating that the stress relaxation follows an exponential law. A slightly more general case may be represented by:

$$\tau = (\tau_0 - \tau_e) \exp\left(-\frac{t}{\lambda}\right) + \tau_e$$

Equation 24

where τ_0 is the value of the stress immediately after the fixing of the deformation and τ_e its limiting-value (residual stress). The quantity λ in these cases is called the relaxation-time. Relations of this nature can be easily obtained for model systems of the kind mentioned at the end of the preceeding section.

It must be observed that few real materials obey the simple laws Equation 23 or Equation 24. In many cases it appears necessary to introduce a “spectrum” of relaxation-times.

47. Experiments in which the rate of shear is kept constant. - The results of such experiments will bear a relation to results found for the course of the deformation as a function of the time for constant stress, as pictured in Figure 1Figure 1 – Standard Experiment

We must expect that when there is decelerating deformation for constant stress, a constant rate of deformation will require a gradually increasing stress.

With accelerating deformations, on the other hand, a constant rate of deformation will lead to decreasing stress.

Finally, when uniformly increasing deformations are reached under constant stress (liquid deformations; compare section 22, sub 3), and experiment in which the rate of deformation is fixed from the start, will lead to a constant value of the stress. In such cases the viscosity-curve

($D - \tau$ −curve) can be deduced just as well from experiments with fixed rates of deformation, as from experiments with fixed stresses.

Phenomena of hardening or softening, or, may be, of thixotropy, can be detected when the same specimen of the material is subjected repeatedly to an experiment in which the rate of shear is always fixed at the same value. Whether the values of the stress to be applied in the second experiment are higher or lower than those applied in the first experiment; and further, what is the influence on this difference of the interval of time between the two experiments, must then be investigated.

48. Experiments with periodically varying deformation. We have seen that, when a fixed value is given to the stress, the establishment of the deformation (or the rate of deformation) in equilibrium with it, will in general require some lapse of time, in consequence of retardation effects or of effects of internal re-adjustment. Re-adjustment phenomena will also lead to changes of the stress, when the deformation, or the rate of deformation, has been fixed.

Whereas it is often difficult to determine satisfactorily what is the most appropriate physical picture for such effects, their detection and measurement can usually be carried out very conveniently by means of experiments in which the substance is subjected to a periodically varying deformation. Either a harmonically varying deformation can be applied by mechanical means, or the stress to which the substance is subjected may be a harmonic function of the time. In both types of experiment a constant value, either of the deformation, or of the stress, may be combined with the harmonically varying part.

When the course of the deformation and that of the stress are recorded simultaneously, the retardation effects will lead to the appearance of a phase lag. In a diagram with stress and deformation as rectangular coordinates, a curve will be obtained showing a loop ("hysteresis loop"). Usually a steady state can be reached in which the same closed curve is continually described. The area of the hysteresis loop is then a measure for the work performed by the stress system on the substance per unit volume in each cycle.

Both the phase lag and the ratio of deformation amplitude to stress amplitude are found to be functions of the frequency applied in the experiment. Measurement of these quantities over an extensive range of frequencies can give important information on the spectrum of retardation-times of the substance, provided its behaviour can be described with the aid of linear differential or integro-differential equations. With an important class of materials (in particular high polymers) satisfactory results are obtained in this way. Usually the description on the basis of a spectrum of retardation-times can also be interpreted in terms of a spectrum of relaxation-times.

A rather extensive terminology has been developed in connection with this subject (Alfrey, Gemant, Kuhn, Leaderman, Simha), which, however, will be left aside here.

The work performed on the material in each cycle is transformed into heat, which must be carried away by conduction or radiation. However, it may be that part of this work is stored in the material through internal changes of structure, .e.g. by the rupture of bonds between atoms or groups of atoms. When this storing of energy and the consequent change of structure is

cumulative, the form of the hysteresis loop may gradually change. Sometimes millions of cycles must occur before these changes become apparent. The change is usually of such a nature that the breaking strength of the material decreases. Its capacity for admitting permanent deformations when subjected to a constant stress is usually diminished and it may become brittle.

These phenomena are described as fatigue: they can lead to fatigue rupture.

Part C: APPLICATIONS

Chapter VII – Examples of the application of the terminology developed in the preceeding sections and comparison with the nomenclature of other authors.

49. The analysis of the types of rheological behaviour, presented in the preceding Chapters, has yielded a system of concepts which can be used as a basis for classification. As a number of these can be grouped together, the question might arise whether they could be combined into something like a decimal system. However, it has been our impression that this would lead to cumbersome and not always convenient formulae.

The following examples may be considered.

Suppose that it has bene found that a certain substance undergoes permanent deformations of appreciable magnitude for all values of the shear stress applied. In this case there is no yield-value.

When a set of flow-curves has been determined, as indicated in Figure 3, let us assume that none of them intersect. An analysis of the set might now reveal that the flow-curves could be divided for example into four groups: the first being formed of curves with a horizontal asymptote, indicating limited deformations; the second representing indefinitely increasing deformations with a decreasing value of $d\gamma_p/dt$, so that there is decreasing flow; the third representing uniform flow; and the fourth accelerating flow. Perhaps it may be possible to determine the threshold values of shear stress which much be exceeded in order to pass from one type to the next

Further analysis might reveal that in the cases represented by the first and second groups, work hardening appears, either reversible or non-reversible.

For values of τ which give rise to uniform flow, a viscosity-curve can be constructed, which may indicate one of the following cases: flow with shear-rate thickening, flow with shear-rate thinning, Bingham flow, or Newtonian flow.

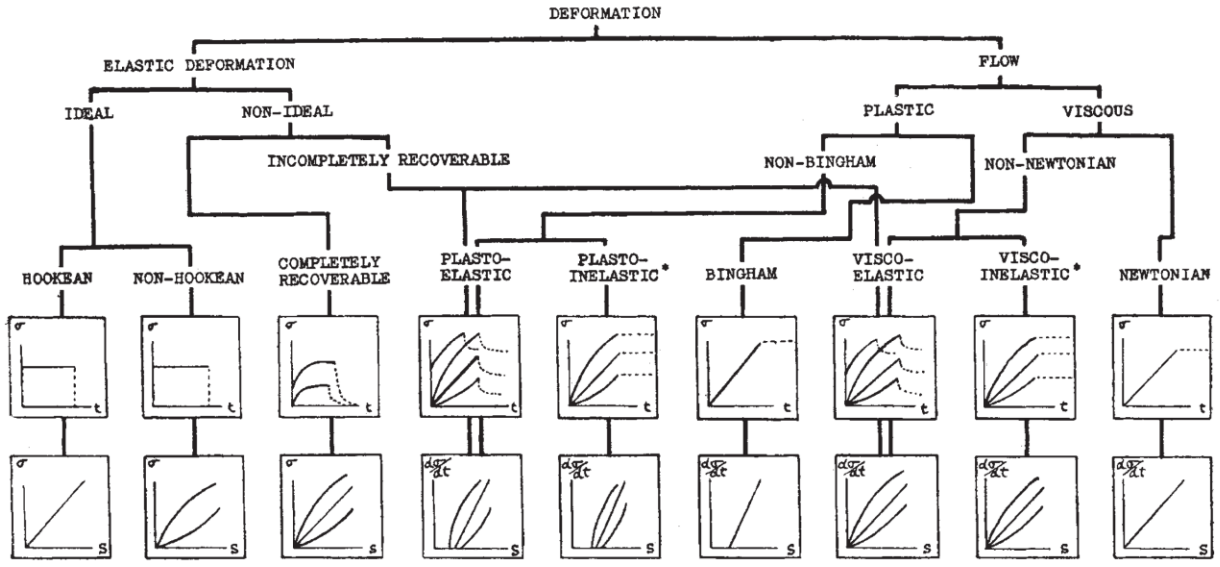
Finally, when accelerating flow is produced, by making τ sufficiently high, thixotropy might appear. Since we had assumed there was no yield-value, our substance behaves as a “thixotropic fluid”.

In the case where deformations are either of limited amount, or decelerating, or uniformly increasing, a certain amount of elastic deformation might be found, - for instance with slow spontaneous recovery -, so that, in this range of τ, there would be fluid-elastic behaviour.

Alternatively the set of flow-curves might show only decelerating flow; or only uniform flow; or decelerating flow for small τ and uniform for large τ; or uniform flow for small and medium values of τ and accelerating flow for very high values, etc.

When a yield-value is found, it would still be possible that much of the behaviour exhibited by the substance for stresses exceeding it would correspond to the description given above. But it would be necessary to give information about the nature of the elastic recovery and the small permanent deformations which may accompany it, for experiments carried out with a shearing stress below the yield value.

50. We shall next consider the scheme of classification, proposed by a Committee of the British Rheologists' Club, published in "Nature" 149, p 702, 1942, and elsewhere, in order to indicate how the cases enumerated in this scheme would be designated according to the terminology of the present Report



(1) σ represents shear strain.
S ,, ,, stress.
t ,, time.

(2) Continuous curves represent deformation under stress.

(3) Dotted curves represent behaviour after removal of stress.

(4) Where two or more curves appear in a diagram, these indicate alternative types of behaviour.

(5) In cases marked * either the σ/t curve or the $\frac{d\sigma}{dt}/S$ curve may be a straight line, but not both, since in this case the systems would be Bingham or Newtonian, which are given separate categories.

(6) The upper series of diagrams in which shear strain is plotted against time represents deformations under constant stress.
The lower series, in which $d\sigma/dt$ is plotted against stress, is for an arbitrary time.

Figure 10

The scheme has been reproduced in Figure 10, with the permission of the Editors of "Nature". It contains two sets of graphs, the upper ones giving the deformation as a function of the time for constant stress, the lower ones giving curves for either deformation or rate of flow as a function of the stress. Where two or more curves appear in a diagram, these indicate alternative types of behaviour. In our present notation, we read γ for σ and τ for $\underline{S}$.

With regard to the main division occurring in the upper part of the Table, we prefer to substitute the term "fluid" for "viscous" and "fluid-elastic", "fluid-inelastic", for "visco-elastic", and "visco-inelastic", respectively.

Coming to the special cases, the first three all refer to what we have called perfectly elastic behaviour, characterised by complete recovery. When the elastic deformations are limited to a few percent, and the modulus of elasticity is large, we may also use the term "perfectly elastic behaviour with rapid spontaneous recovery", which can also be designated as "ideally elastic behaviour" (or, as the case may be, "ideal solid beahvior"). In case three there is "perfectly elastic behaviour with slow spontaneous recovery".

Cases where recovery is obtained only on stimulation, have not been considered in this scheme.

In case 1, the elastic behaviour is Hookean; in case 2, non-Hookean, the modulus of elasticity either increasing or decreasing when the deformation becomes large.

In the upper picture of case 3 different forms of retardation effects appearing after the application of the stress, have been indicated. The lower picture again indicates Hookean and two forms of non-Hookean behaviour.

51. In all other cases (4-9) of the Table of the British Rheologists' Club, permanent deformations make their appearance. As mentioned, the curves given in the upper diagram refer to the course of the total deformation r considered as a function of $\underline{t'}$.and thus correspond to the curve in our Figure 1Figure 1 – Standard Experiment and Figure 2. There are no curves given for the course of $\gamma_p$as a function of the duration $\underline{t}$ of the experiment.

The curves show that in cases 4 and 7 there is some recovery, so that the permanent deformations are accompanied by elastic deformations, whereas in cases 5, 6, 8, 9 this is not the case.

We may further assume that the trend of these curves gives a sufficient indication as to whether there is decelerating, uniform or accelerating flow, although a set of flow-curves ($\gamma_P - t$-curves) for a series of values of τ (as given in our Figure 3) would furnish more precise information. Such a set of flow-curves might reveal for example, whether a substance would show more than one of these types of behaviour, depending on the magnitude of the stress.

There is the further possibility that permanent deformations of limited magnitude might be found for certain values of the stress, a case not indicated in the scheme.

The curves given in some of the lower diagrams can be compared with our $D - \tau$- curves only in those cases where the curve in the upper diagram, for a certain domain of values of τ, approximates to a straight line, indicating uniform flow. It appears that originally the curves of the lower diagrams were meant to give the value of $d\gamma/dt$ taken at an arbitrary (fixed) value of t, plotted against τ. A representation of such data, however, cannot be compared with the scheme adopted in the present Report.

If we assume that uniform flow is present, the following alternatives can be distinguished: flow with shear-rate thickening (curve concave towards the stress axis in cases 4,5,7,8); Bingham flow (straight line in cases 4,5,6); Newtonian flow (straight line through the origin in cases 7,8,9); flow with shear-rate thinning (curve concave towards the rate of flow axis in cases 4,5,7,8).

It must be observed that one of the curves given in the upper diagram of case 4, and likewise one in the upper diagram of case 7, indicates the presence of elastic deformations. Formally our definition of a "Newtonian liquid", as given in section 25 sub (B), should always be applicable to case 7, when the $D - \tau$ −curve is straight. In practice it might perhaps be preferred to exclude liquids with conspicuous elasticity. In the cases 4-6 it is indicated that uniform flow starts only if the stress exceeds a certain threshold, whereas in cases 7-9 uniform flow starts from zero stress. Hence in these latter cases (7-9) there is fluid flow (either fluid-elastic, or fluid-inelastic). In cases 4-6 more information is necessary. We must know what happens if the stress is below the threshold: do we then find elastic deformation only, or are there still permanent deformations, though of limited or of decelerating character? It is only in the first case that the intersection of the $(d\gamma_p/dt) - \tau$-curve with the horizontal axis gives the true yield-value. In the second case it gives no more than the threshold value for uniform flow, but the actual yield-value will be lower and might even be zero. The term "plastic" can be used only if this point is made clear.

52. We will next consider a classification proposed by J.Pryce-Jones in "The flow of suspensions thixotropy and dilatancy", Proc. Univers. of Durham Philos. Soc.10 (1947), pp.446-448.

The substances of Pryce-Jones' Class A are characterised as having "viscosities independent of the rate of shear as well of time of rest after stirring has ceased". We must assume that there appears uniform flow, so that a viscosity indeed may be defined. This viscosity then appears to be independent of the stress (or of the rate of shear). When a repeated experiment is carried out, no change of viscosity is found. These substances will consequently be Newtonian liquids.

The substances of Class B present a "viscosity which is a function of the rate of shear, and decreases with increasing rate of shear, but is independent of time". We again assume that they present uniform flow, in order that a viscosity can be defined. This viscosity appears to be a function of the rate of shear (or of the stress). We assume that there is no yield-value. We are then dealing with non-Newtonian liquids, showing shear-rate thinning. (If a yield-value were present we would have plastic substances admitting uniform flow with shear-rate thinning). When a repeated experiment is performed, no change of viscosity is found.

Concerning Class C Pryce-Jones writes: “The viscosities of these systems immediately after stirring are practically independent of the rate of shear; in other words, they are almost true Newtonian liquids. When left at rest for increasing periods of time, their viscosities increase as indicated by the vale of the maximum deflections. As shearing continues the gel gradually breaks down and the deflection decreases to the value observed when the experiment is made immediately after stirring has ceased”

We must interpret this description as indicating that a repeated experiment shows a decreased resistance to flow, while there is a gradual spontaneous return to the original state, when the substance is left to itself. This is an indication of thixotropy. The point of rheological importance is whether there can be found a yield-value in the unstirred state, and whether this yield-value disappears after sufficient stirring. If this appears to be the case, we think it appropriate to speak of a “thixotropic plasic-fluid substance”. Pryce-Jones considers this as an indication of a gel/sol/gel transformation, as he defined a gel as any system showing even the smallest rigidity and as a sol any system which does not (i.e. our “fluid”).

Concerning Class D Pryce-Jones states that “all members of this class possess anomalous viscosity immediately after stirring has ceased. Further their viscosities rapidly increase when left at rest for even a short interval of time, but this rate of increase does not continue with increasing intervals of rest. This type corresponds to False-Body, as described in my earlier papers. Whereas Class C appears to be a true gel/sol/gel transformation induced by shear and subsequent rest, i.e. true thixotropy, Class D appears to be a partial dispersion of a gel under the influence of shear which, in my mind should not be regarded as true thixotropy.”

Our comment is that from the rheological point of view the difference between classes C and D is not quite clear. The main point seems to be that with substances of class D there is a very quick return to the original state. If, however, we may assume that with these substances a yield-value can be found also after stirring, but lower than the original yield-value we would be dealing with what we should call “thixotropic plastic substances”.

The description of Class E as given by Pryce-Jones runs: “The members of this class are dilatant fluids and the viscosity at the lowest rate of shear is independent of time. It should be realised, however, that when the critical rate of shear has been reached, that is the rate of shear at which the viscosity begins to increase, it has been found that continued shearing at a constant rate of shear leads to an increase in viscosity with time.”

Again this description does not give sufficient information from the rheological point of view. We need data concerning the form of $\gamma_p - t$- curves. It must then be decided, (a) whether there is an intersection of these curves, (b) whether there is approach to regions with limited deformations, or with decelerating, uniform, or accelerating flow, or whether all such cases are present. Only on the bases of such data will it be possible to indicate which case is represented by these substances.

53. Mr. J. Pryce-Jones has given a further list of cases in his contribution to the discussion on nomenclature at the International Rheological Congress of September 1948 at Scheveningen.

The following types of substances are mentioned (we give, first Mr. Pryce-Jones' descriptions, and then add our comments);

Newtonian liquids. "Systems with viscosity independent of time or rate of shear". We must consider these to be substances which flow uniformly and thus possess a viscosity independent of stress (or of the rate of shear), while a repeated experiment does not show a change of state.

Non-Newtonian liquids. "Systems with viscosities dependent on the rate of shear but independent of time of shear". – We must assume the same properties as before, but for the fact that the viscosity is a function of the rate of shear (or of the stress).

Plastic dispersions. "Dispersions possessing a finite yield-value; their viscosities will depend on rate of shear, but are independent of time of shear". – We must apparently assume that once the yield-value is exceeded, there is uniform flow (though this might be otherwise), so that a viscosity can be defined. This viscosity is a function of the rate of shear (or, of the stress), but it does not change when a repeated experiment is performed. We should call these systems "plastic substances exhibiting uniform flow".

Pseudo-plastic dispersions. "Dispersions possessing no yield-value; their viscosities depend on the rate of shear, but are independent of time." – Since we again assume that these substances undergo uniform flow, so that a viscosity can be defined, they are simply non-Newtonian liquids. If instead of uniform flow under constant shear stress there were limited flow, decelerating flow, or perhaps accelerating flow, we ought to characterise their behaviour by applying the more general term "fluids".

Dilatant dispersions. "Dispersions which are Newtonian liquids at low rates of shear, but their viscosities increase with increasing rate of shear above a minimum critical value. The value of the viscosity is independent of time." – From this description it is not sufficiently clear what is meant. These substances may be fluids or liquids exhibiting shear-rate thickening; in certain cases the extreme form of shear-rate thickening may occur which leads to a decrease of $d\gamma_p/dt$ with increased values of τ , or even there could be found shear-rate blockage (section 28)

Shear-hardening dispersions. "Dispersions which increase in viscosity at constant rate of shear with increasing time of shear, but revert to their original viscosity when left at rest." – This definition is inadequate. It is possible that there be decelerating flow, in which case no viscosity can be defined. It is also possible that uniform flow is obtained, but that a repeated experiment shows the internal state of the substance to have changes, in such a way that a big resistance to deformation is offered. Only in such a case we have to do with shear-hardening. When there is a return to the original rheological properties when left at rest, there is "reversible shear hardening".

Thixotropic dispersions. "Dispersions which show the internal reversible gel/sol/gel transformation. In a state of rest they are gels which possess a yield-value and their viscosities are a function of the rate of shear, as well as the time of shear and time of rest. At high rates of shear, they are sols, their viscosities in the sol state are independent of the rate of shear or the

time of shearing". – According to this description these substances possess a yield-value in the state of rest, whereas there would seem to be no yield-value after stirring. As there is a return to the original state when left at rest, these systems are "thixotropic plastic-fluid substances".

Sol. "A colloidal system which possesses no yield-value, its viscosity is independent of the rate of shear or time of shear, in other words, a Newtonian liquid". – We must observe that we are certain that this will not apply to all those systems commonly called sols.

Gel. "A colloidal system which possesses a yield-value. Its viscosity is a function of the rate of shear. If the gel is thixotropic or false-bodied the viscosity is a function of the time of shear, or the time of rest, as well as the rate of shear". – This is therefore a plastic material, which may exhibit thixotropy either fully, or incompletely. We believe that the definition of a gel as given by Pryce-Jones will include many materials which have not hitherto been regarded as gels

False-Body dispersions. "Dispersions which are permanent gels, but the viscosity of the gel is a function of the rate of shear as well as the time of shear or of rest". – They may correspond to what we have called "thixotropic plastic substances".

Rheopectic dispersions. "Thixotropic or False-body dispersions in which the rate of increase of viscosity is accelerated by gentle rhythmic shearing".

We would prefer "consistency" to "viscosity" in many of the above definitions

54. Some views on the thixotropy and related forms of rheological behaviour have been developed
by H. Green and Miss. R. N. Weltmann (Colloid Chemistry Vol. VI, p. 328, 1946).

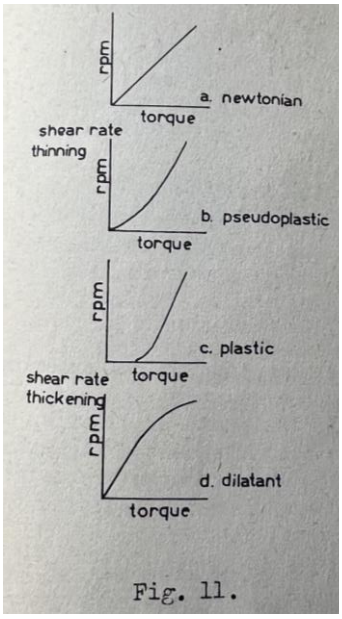


Figure 11

In their Figure 1. (p.330), of which the principles are shown in our Figure 11, schematic "consistency curves" are pictures for different types of suspensions, giving the number of revolutions per minute of a rotating viscometer (ordinates) as a function of torque (abscissae).. They consequently correspond to our $D - \tau$ – curves, but it should never be forgotten that such curves can only have a physical sense when it has been ascertained that uniform flow with constant rate D (i.e., with a constant number of revolutions per minute) is actually obtained for every value of the shear stresses or of the torque in the range considered. This is certainly not the case with many thixotropic systems

The first curve is a straight line through the origin, characterising Newtonian flow.

The second curve starts from zero torque, to that there is no yield-value. The curve has its concave side toward the vertical axis. Systems exhibiting such behaviour are called "pseudo-plastic" by the authors. In our terminology we prefer to replace this term by "liquid flow with shear-rate thinning"

In the third case a curve of similar nature is pictures, but starting from a point on the horizontal axis, corresponding to a finite intercept on the τ-axis. We may assume that this indicated a yield-value, although it might be only threshold value for the appearance of uniform flow, so that the true yield-value (below which no permanent deformations are obtained of magnitude comparable with the elastic ones) might be lower. As the authors mention, there is thus plastic flow, but we would qualify this behaviour more precisely as "uniform plastic flow with shear-rate thinning".

In the fourth case the curve again starts from the origin, but this time it has its concave side to the horizontal axis. The authors call this behaviour "dilatant". We consider this term inappropriate and replace it by "liquid flow with shear-rate thickening"

55. In a volume on "The Principles of Rheological Measurement" (1946), to be published by the British Rheologists' Club, H. Green has given some very useful observations on the principles involved in the measurement of thixotropic behaviour (p.75). In the paper quoted in the preceding section, however, Green and Miss Weltmann describe how thixotropy may be detected, by taking "up-and-down-curves" with a rotational viscometer, which curves give a relation between angular velocity (indicating rate of shear) and torque (indicating shear stress). It has already been mentioned (section 41) that, while this method can give useful qualitative information on the hysteresis characteristic of thixotropy, it does not give us quantitative data of definite physical value.

The authors attempt to develop their method of observation into a scheme for quantitative analysis by introducing certain coefficients which play a part in a set of equations, deduced partly in this and partly in other papers (e.g. Ind. And Eng. Chem. 18, p.167, 1946). However the concepts upon which these equations are based are somewhat vague.

The quantity U introduced in their papers and called "plastic viscosity", can have a meaning only when calculated from a pure $D-\tau$ – curve, obtained from measurements in which it has been ascertained that uniform flow was present for every value of τ applied. It is then equivalent to the derivative $d\tau$/dD in our notation. Likewise the "intercept" (f) obtained when the $D-\tau$ – curve is extended downwards (more or less hypothetically) until it cuts the τ-axis, can have a meaning only for a true $D-\tau$ – curve.

The coefficient M, defined as the "loss in shearing force per unit are per unit increase in velocity gradient", is related to the second derivative $d^2\tau/dD^2$, but its definition is unsatisfactory. As indicated by the formula for the second derivative, it must contain the square of the increase of the rate of shear in the denominator, since otherwise it does not possess a definite limiting value when this increase is reduced to zero.

The concepts mentioned thus far refer to the properties of the $D-\tau$ – diagram, and do not in themselves contain any reference to thixotropy.

The next coefficient (B) however, does: this coefficient measures the decrease of U ("the thixotropic breakdown") with time, and is defined by:

$$B = t\frac{dU}{dt} = \frac{dU}{d(\ln(t))},$$

where t is the time of "agitation" (shearing in some particular experiment at a certain rate). A proper measurement of such a quantity is possible only provided that a $D - \tau$ – curve can be found in the state produced by the agitation, without introducing further serious disturbances of structure. The authors state that this is the case when the measurements are performed at decreasing rates of shear, below the rate applied in producing the breakdown.

It seems that a decrease in U is accompanied by an increase of the intercept f on the τ-axis. A coefficient is defined by the equation

$$V = -df/dU.$$

It would be important to have a quantity measuring the time needed to return to the original state.

56. We will now consider a classification proposed by Sir Charles Goodeve and mentioned by him in the discussion at the Rheological Congress. This classification begins by distinguishing between "solids" and "fluids" or "liquids" (no distinction is introduced between the latter two terms).

"Solids" are characterised by their ideal or approximate elastic behaviour. The terms "true" or "ideal solids" and "ordinary solids" are easily understood without further comment. The term "rubber-like solids" is mentioned without clear definition: it seems to refer to substances whose elastic behaviour shows appreciable retardation, and for which experiments with oscillating stresses reveal a hysteresis-loop.

When an elastic limit is present, which term can be considered as equivalent to a yield-value, Goodeve distinguishes "brittle solids" and "plastic solids", according to whether fractur or plastic flow occurs above the elastic limit. While the distinction itself is clear, we must observe that, taken in this sense, all substances will show a limit to their elastic behaviour, since no substance can withstand infinite stress.

In the case of plastic flow a distinction is made between "simple plastic flow" and "complex plastic flow". It is assumed that in the first sub-case the stress in the plastic range is independent of both the magnitude of the deformation and of the rate of deformation. A substance falling in this case apparently cannot withstand a stress larger than the yield-stress. In mathematical theories of behaviour in the plastic range this supposition gives a useful approximation, and an ideal body behaving in such a way is called a "Saint-Venant body" by Reiner (compare section 6363 sub (V)). We would, however, not take it as the standard specification of plasticity, which is a much more general concept.

In the case of "complex plastic flow" the term "Bingham solids" is used when the stress increases with the rate of deformation. This apparently refers to plastic substances which exhibit uniform flow under constant stress. In contrast with these "Bingham solids", substances showing work hardening are mentioned. From what has been said in Chapter III and Chapter VChapter

V – Application of repeated experiments performed on a single specimen. the reader will see that many types of behaviour other than uniform flow are possible with plastic substances, while the strict definition of work hardening can be based only on the outcome of repeated experiments. Goodeve finally mentions "creep", as a slow continuous increase of deformation with stress below the recognised "elastic limit of stress", that is, below what we call the yield-value. Probably Goodeve's "creep" may be considered as equivalent with our micro-flow of the first kind (sections 22, 23).

57. Coming to the second half of Goodeve's scheme, referring to fluids and liquids, a first distinction is made according to whether elastic deformations accompany flow, or are absent. In the first case the terms "elastic fluids" and "elastic liquids" are used; in the second case the term "true fluids" is applied. We had not introduced these terms, although they might be included. Instead of "true fluids" we should prefer "inelastic fluids" or "ordinary fluids".

"Viscous flow" is then considered, meaning apparently the appearing of uniform flow with constant stress. Cases with non-uniform flow are not mentioned, a serious omission. The usual distinction is made between Newtonian and non-Newtonian liquids. In the case of non-Newtonian liquids, the subcases of viscosity decreasing with increasing rate of shear, and viscosity increasing with increasing rate of shear are distinguished.

The term "thixotropic" and "thixopetic" liquids, respectively, are proposed. These terms, however, are at variance with our definition of the concept of thixotropy, which was based upon the behaviour in repeated experiments. Goodeve in his scheme mentions something about "influence of previous history" and "reversibility" (slow or quick), but the distinctions made here are not sufficiently worked out.

It is finally mentioned that highly viscous fluids, such as glasses, may behave as brittle solids for short term applications of stress, and are highly viscous Newtonian (or, possibly non-Newtonian) fluids for long term applications.

58. Dr. H. Leaderman of the National Bureau of Standards, Washington D.C., has prepared a memorandum on Rheological Nomenclature for Elastic and Inelastic Behaviour of High Polymers, presented at the meeting of the (American) Society of Rheology, New York, November 6, 1948, and distributed to interested rheologists, of which he has kindly forwarded us a copy. The proposed terminology is summarised at the end of the memorandum. In the following lines we shall compare some of Leaderman's terms with ours.

Under the heading "Elastic Behaviour of Polymer Systems", the terms for "ideally elastic behaviour" and "Hookean elastic behaviour" are defined in a way corresponding to the concepts introduced in our sections 32 and 30. A distinction between "ordinary elasticity" and "rubber-like elasticity" is made to depend on the ratio of the shear modulus to the bulk modulus. This distinction is thus independent of that between ordinary and high elasticity given in section 30.

A material is said to exhibit "visco-elastic behaviour" if it manifests Hookean behaviour in shear when the time-scale of the experiment is either very small or very long compared with some characteristic time, and when the shear moduli for very rapid and very slow experiments

are different, the former being much larger than the latter. In order to illustrate this concept the following differential equation is considered as an example of a specialised type of "visco-elastic behaviour"

$$\left(\frac{G_1 + G_2}{G_2} + \frac{\eta_1}{G_2}\frac{d}{dt}\right)\tau = \left(G_1 + \eta_1 \frac{d}{dt}\right)\gamma$$

Equation 25

A system which behaves according to this equation shows a modulus G_2 for very rapid deformation, and a modulus $G_1 G_2/(G_1 + G_2)$ for very slow deformation. As a system of this kind will show complete recovery, it would come under the class of perfectly elastic behaviour of our terminology. The concept of two distinct elastic moduli is one belonging to the theory of systems with a spectrum of retardation- or relaxation0times, to which reference was made in section 48.

Leaderman's concept of "visco-elastic behaviour" cannot be compared with the concepts "fluid-elastic" and "liquid-elastic" introduced in our section 31.

Leaderman adds that in many papers the term "visco-elastic behaviour" is connected with a form of behaviour described by Equation 26 to be given below rather than with that described by Equation 25.

Leaderman develops a number of concepts in relation to the equation just given above, e.g.: retardation-time, retarded elastic deformation, retarded recovery etc. Attention is given to stress relaxation and to the relaxation-time. The quantity η_1 is called the "internal viscosity", as distinct from another kind of viscosity, to be mentioned later

The term "creep" (previously called by Leaderman "primary creep") is used to denote the increase in elastic deformation with time under constant stress. Retarded elastic recovery is called "creep recovery".

Much attention is given to the behaviour of a substance following Equation 25 under the action of a harmonically varying stress; and a number of terms e.g. "complex shear modulus" etc., are defined. As they are rather specialised and as their application presupposes the validity of the differential equation, we will not repeat them here.

59. Under the heading "Inelastic Behaviour of Typical Systems" Leaderman considers systems in which the deformation does not entirely disappear after removal of the stress

The term "inelastic deformation" is used to denote the non-recoverable component of the deformation, and thus corresponds to our permanent deformation γ_p. A distinction is made between "non-yield-point behaviour" and "yield-point behaviour".

For non-yield-point behaviour the relation between shear stress and shear is represented as a first approximation for small values of the stress by a differential equation of the form

$$\left(G_1 \frac{d}{dt} + \eta_1 \frac{d^2}{dt^2}\right)\gamma = \left(\frac{G_1}{\eta_2} + \left(\frac{\eta_1}{\eta_2} + \frac{G_1 + G_2}{G_2}\right)\frac{d}{dt} + \frac{\eta_1}{G_2}\frac{d^2}{dt^2}\right)\tau$$

Equation 26

Under the action of a constant stress there will finally be obtained an inelastic deformation increasing linearly with time. This type of inelastic deformation is denoted by the term "flow". The concept of flow introduced in this way does not greatly differ from our definition, although we have not restricted it to uniform increase of the magnitude of the deformation.

"Newtonian flow" (or "viscous flow") is defined by proportionality between rate of flow and shear stress. The coefficient η_2 is called the "principal viscosity" or "melt viscosity" (or simple "viscosity), in contra-distinction to the "internal viscosity" η_1.

In discussing systems showing yield-point behaviour attention is called to the special case when the inelastic deformation increases proportionally to time and to the excess of the stress over the yield-value, in which case the material is said to exhibit "plastic flow" or "Bingham behaviour". It is observed that in general the inelastic deformation in polymeric systems does not follow Bingham behaviour, and that work hardening appears.

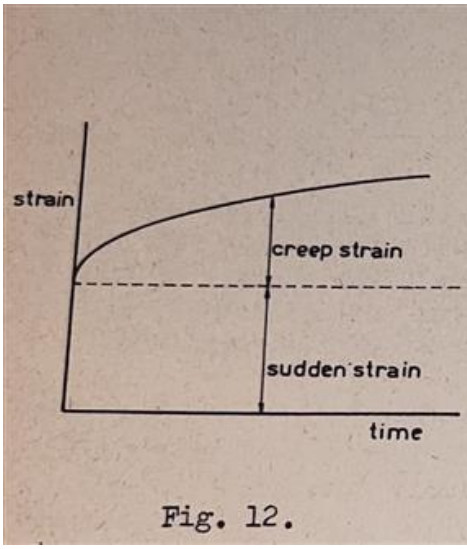
A few words are added concerning stimulated recovery (swelling recovery and thermorecovery are mentioned), for which we had proposed the term "redress" (see section 29).

Finally it is observed that materials such as polystyrene may exhibit yield-point behaviour in compression, while in tension they behave as brittle glasses.

60. Creep. – In the literature on the rheological behaviour of metals, glass, filamentous materials, and other substances, the term "creep" is of frequent occurrence. It is used mostly in connection with extension experiments, but it has a general meaning, so that the term can be used also in relation to shear.

It is difficult to find a precise definition of creep. On the whole it seems to refer to the increase of the total deformation without distinguishing between the recoverable and the non-recoverable parts. This make it less satisfactory as a basis for a precise analysis of the underlying physical phenomena.

By way of example, E. Orowan, in a paper on “The Creep of Metals” (West of Scotland Iron and Steel Institute, 1947), starts from what he calls “the traditional description of plastic properties, as represented by the stress-strain curve”. Without going into those details which



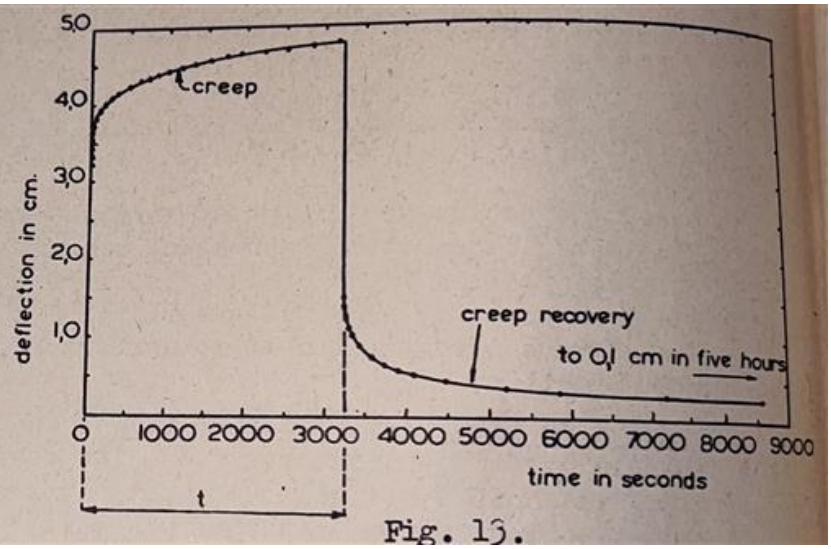


Figure 12 *Figure 13*

must be observed in order to obtain such a curve in exact form, Orowan immediately passes on to strain-time curves and states that in any tension test, rapid application or increase of stress produces a rapid extension, which, however, does not stop suddenly, but decreases gradually in velocity and goes over into a slow plastic flow which is called “creep” (Figure 2. Of his paper, see our Figure 13). Some particular forms of creep are then mentioned, as “quasi-viscous creep”, “transient creep” etc.

No sharp distinction is made between permanent and elastic deformation and on the whole creep seems to be taken as a combination of a recoverable and a non-recoverable deformation, the later being of a magnitude comparable with (or usually greater than) the former. The introduction of the term “creep recovery” in a further part of the paper indeed shows that in Orowan’s picture both types are combined.

With regard to the terms “transient creep” and “quasi-viscous creep”, which Orowan uses in particular connection with the well-known experiments of Andrade on the creep of metals, it would appear that on the whole (at least in the case of these experiments) these concepts refer to phenomena of flow, i.e. to increase of the permanent deformation with increase of the length of the period of action of the force. “Quasi-viscous creep” seems to correspond approximately to what we would call uniform flow. “Transient creep” has a rate vanishing with time. It is apparently defined by the curve which is obtained when the uniform flow, extrapolated back wards to the beginning of the experiment, is deducted from the curve for the total deformation, observed as a function of the time $\underline{t'}$ during the action of the stress. The “transient creep” might still include some effects of retarded elastic deformation, together with a component of true non-uniform flow.

Leaderman, in his book "Elastic and creep properties of filamentous materials" (Washington D. C., 1943), again without giving a precisely framed definition of creep, begins by introducing a term "primary creep" as synonymous with "delayed elasticity". Afterwards a term "secondary creep" is used to indicate an additional "non-recoverable creep", which is also called "mechanical conditioning". Apparently an attempt is made here to separate the non-recoverable from the recoverable part of the deformation. In the paper referred to in sections 58 and 59, Leaderman uses the term creep without further qualification, for what he formally called "primary creep"

We would recommend limiting the meaning of the term "creep" to the sense of Leaderman's "primary creep" viz. as the (gradual) increase with time of the elastic part of the deformation, while the term "creep recovery" could be used for the gradual decrease of an elastic deformation after the force has been taken away. The term "creep" would then refer to the retarded part of the elastic deformation as contrasted with the part appearing rapidly (or disappearing rapidly after the removal of the force).

Since we use the term "flow" to indicate the increase with time of the non-recoverable part of a deformation, and micro-flow when this increase is very slow, we would then have a clear distinction between the concept of creep and that of flow.

When for practical purposes the term creep is nevertheless used with reference to the behaviour of the total deformation, it should be clearly state what is meant. The terms "apparent creep" or better still "apparent flow" or "apparent micro-flow" are recommended, as indicated already in 27.

Two examples may still be taken from existing literature to bring to evidence the difficulties resulting from lack of sufficient data.

T'ing – Siu Kê, of the school of C. Zener, in a publication entitled "Experimental evidence of the viscous behaviour of grain boundaries in metals", Phys. Review 71, p. 542, 1947, gives a curve shown in our Figure 13. The period t during which the load acted on the specimen was about 3200 seconds. The decrease of the deformation after the removal of the load gradually becomes extremely slow. It may be assumed that the ordinate corresponding to γ_p finally falls to about 0.1 cm and as this appears to be a very small fraction of the total deformation, the applied stress must have been below the yield-value. The diagram, which gives no more that a single point of the $\gamma_p - t$ – curve for the applied stress, cannot afford information concerning micro-flow. But it shows very clearly the retardation effects, both in the establishment of the elastic deformation and in the recovery phase, and the terms "creep" and "creep recovery" inserted by the author along wit the two branches of the curve, are in accordance with what was proposed by Leaderman.

B. Chalmers (Proc. Roy. Soc. London A.156, p 427, 1936) has described fairly distinct regimes of creep, which he distinguishes as "micro-creep" and "macro-creep" respectively. There is a fairly well defined value of the critical stress marking the transition from one regime ot the other. We will suppose that in this case the observed deformations are non-recoverable. Then the observed increase should be denoted as fellow. Since the rate of increase in the first regime is

below 10^8 sec^{-1} , so that the deformations even for a period of several hours remain very small, we may assume that we are below the yield value in the sense as defined in 20, and that we have to do with micro-flow of the first kind.

61. In his book "Die Viskosität der Kolloide" (Dresden und Leipzig, 1942) W. Philippoff has introduced some definitions which can be compared with the nomenclature of the present Report. Philippoff starts (p.30) by defining Rheology or "Fleisskunde" as the theoretical and experimental investigation of flow (the behaviour under the action of shear stress) of all kinds of materials. At p.33, however, he restricts his treatment to "laminar stationary flow", which apparently implies that he is concerned only with what we have called uniform flow, in which the rate of flow, $d\gamma_p/dt$, for constant stress τ, has or ultimately attains a constant value. Philippoff is of the opinion that rheological treatment will usually become impossible when "laminar flow" is not obtained.

The book consequently refers to such properties as can be identified from $D-\tau$ – diagrams, as described in our section 25. We have called the curve, given by such a diagram, a "rate of flow-stress-curve" or "viscosity-curve". Philippoff uses the term "Fleisskurve" for it. Since we have taken the word "flow" in a more general sense than has Philippoff, so that it embraces also all cases of non-uniform flow, we believe that the term "flow-curves" is not a proper name for the $D-\tau$ – curve. The term "flow-curve" has been used by us, on the contrary, as the general names for the curves giving the course of γ_p as a function of t, i.e. the curve spicturing the flow of any substance in its most general aspect. We observe that if the term "Fliesskunde" us accepted as the German translation for "rheology" (see above), the meaning of the German "Fliess" must be extended to include all types of deformation, as has been done by the international usage for the Greek "$\rho\epsilon\omega$".

Viscosity is defined by Philippoff as the ratio τ/D, in conformity with our Equation 15, section 26. Philippoff , however, uses the term "apparent viscosity" (denoted by η'), when it is desired to mark the difference from the ordinary viscosity of Newtonian liquids. This term thus differs from our "apparent viscosity" as introduced in section 27; it corresponds rather to what we have called "anomalous viscosity" (section 26). The term "Strukturviskosität" is sometimes used to describe the relation between either η' and τ or η' and D. We would discourage the use of this term (see section 25)

Particular attention is given to η_0 ("Anfangsviskosität"), the viscosity found with very small values of τ and D; and to η_∞, the limiting value of the viscosity for very large values of τ. It is assumed that the latter limit exists, at least for many substances showing uniform flow.

Plasticity is mentioned in connection with the $D-\tau$ – diagram: it is stated that plastic substances do not exhibit flow so long as τ remains below a certain threshold value ("Fliessgrenze"). As no reference is made to non-uniform flow, Philippoff's definition (in which it seems to be assumed that uniform flow appears as soon as τ is exceeds the "Fliessgrenze"), is less complete than ours.

Philippoff also pays some attention to the fact that an observed "Strukturviskosität" may depend on the treatment to which the material has been subjected. This will be the case with concentrated suspensions with "zerstörte Gallerte" (stirred and ruptured gels). Apparently this refers to the phenomena of thixotropy. It must be observed, however, that in general Philippoff omits to give precise definitions. Terms such as "Strukturviskosität" (p.4, 113), plasticity (p.38), thixotropy (p.164), dilatancy (ibidem) suddenly turn up without further comment. They are apparently taken over as known, leaving it to the reader to find out for himself in what sense they should be accepted.

62. Remark on Nutting's and Scott Blair's equations. It has been observed by various authors (see G. W. Scott Blair's lecture before the Rheological Congress 1948. P. I-98) that the behaviour of a large group of plastics obeys an equation of the form

$$\psi = S^{\beta}\sigma^{-1}t^{k}$$

Equation 27

where S is the stress, σ the strain, t the time, while $\psi, \beta,$ and k are constants. This equation was originally proposed by P. G. Nutting in a paper "A new general law of deformation", Journ. Frankl. Institute 191, p.679, 1921. The equivalent form for shear would be, in our notation:

$$\psi = \tau^{\beta}\gamma^{-1}(t')^{k}$$

Equation 28a

where we have written t' to indicate the continually increasing time in an experiment in which τ is kept constant. By γ is meant the total deformation.

The equation thus describes a set of $\gamma - t'$ - curves or "apparent flow-curves", according to the terminology of section 27.

A similar remark applies to another general equation, proposed by Scott Blair for the cases where Nutting's equation is inadequate:

$$\frac{d^{\mu+\omega}}{dt'^{\mu}\tau^{\omega}} = \text{Constant}$$

Equation 298

where μ and ω are fractional.

These equations have also been used for the study of stress relaxation at constant deformation

Scott Blair's treatment depends on what he calls "the Principle of Intermediacy". It is not analytical and deformations are therefore not divided into recoverable and non-recoverable parts. These equations thus fall somewhat outside the system of classification developed here. They belong in a category similar to the concept of "creep", when this term is used for the slow

increase of total deformation without distinguishing between its recoverable and non-recoverable components.

63. Reiner's standard types of rheological behaviour. Reiner has introduced a number of idealised types of rheological behaviour, which can be described in relatively simple terms and serve as standards of comparison in the analysis of the behaviour of any actual substance. The behaviour of these standard types can be expressed by means of linear relations (see Rheology Bulletin 16. No. 3, 1945 or Journ. Scientific Instruments 22, p. 125, 1945).

The simplest standard types, already introduced before Reiner's publications, are:

(I) The Hookean Solid, a perfectly and ideally elastic solid (see section 32) with Hookean elastic behaviour (section 30) governed by the equations

$$\tau = G\gamma_e;\ \gamma_p = 0$$

Equation 18, Section 30

(II) The Newtonian liquid (section 25), governed by the equation

$$\tau = \eta \frac{d\gamma_p}{dt} = \eta D; \gamma_e = 0$$

Equation 15, Section 25

To these Reiner adds:

(III) The Kelvin solid, governed by the equation

$$\tau = G\gamma_e + \eta_K \frac{d\gamma_e}{dt}$$

Equation 19, Section 34

and thus showing perfect elasticity accompanied by retardation. The coefficient η_K is called by Reiner: "solid viscosity"

(IV) The Maxwell liquid, governed by

$$\frac{d\gamma_p}{dt} = \frac{\tau}{\eta} + \frac{1}{G_M}\frac{d\tau}{dt}$$

Equation 21, Section 34

where η is the ordinary viscosity equal to λG in Equation 21, while G_M is a modulus of "liquid elasticity" , determining an elastic deformation, accompanying liquid flow.

The Kelvin solid can be obtained by parallel coupling of a Hookean solid and a Newtonian liquid:

$$\mathrm{K} = \mathrm{H} \mid \mathrm{N}\ ;$$

whereas the Maxwell liquid is obtained by coupling in series of these systems

$$\mathrm{M} = \mathrm{H} - \mathrm{N}.$$

More complicated systems can be obtained by repeated application of parallel couplings and couplings in series.

(V) Furter, a Saint-Venant body is defined, which is characterised by the property that no permanent deformation appears so long as the stress is below the yield-value, whereas as soon as this point is reached, a permanent deformation appears, which can reach any value, while a part of the stress equal to the yield-value remains balanced by internal reactions in the body.

The Saint-Venant body can be combined with the others, e.g.:

(VI) A Bingham body is obtained by coupling a Saint-Venant body and a Newtonian Liquid in series:

$$\mathrm{B} = \mathrm{St}\,\mathrm{V} - \mathrm{N};$$

its behaviour is governed by the equation:

$$\frac{d\gamma_p}{dt} = \begin{cases} 0, & (\tau < \tau_f) \\ \dfrac{(\tau - \tau_f)}{B}, & (\tau > \tau_f) \end{cases}$$

Equation 30

(VII) A Schwedoff body is obtained by coupling a Bingham body and a Hookean solid in series.

$$\mathrm{Schw} = \mathrm{B} - \mathrm{H} = \mathrm{St} - \mathrm{N} - \mathrm{H}$$

its equation is

$$\frac{d\gamma}{dt} = \begin{cases} \dfrac{1}{G}\dfrac{d\tau}{dt}, & (\tau < \tau_f) \\ \dfrac{(\tau - \tau_f)}{B} + \dfrac{1}{G}\dfrac{d\tau}{dt}, & (\tau > \tau_f) \end{cases}$$

Equation 3031

(VIII) A Schofield-Scott Blair body is defined by the equation:

$$\mathrm{Sch.\,ScB} = \mathrm{N} - \mathrm{H} - \mathrm{StV} - \mathrm{K} = \mathrm{M} - \mathrm{StV} - \mathrm{K} = \mathrm{Sch} - \mathrm{K}$$

By coupling two Maxwell liquids in series another Maxwell liquid is obtained. Parallel coupling of two Maxwell liquids, however leads to a more complicated liquid, characterised by more than one relaxation time.

APPENDIX

VISCOSITY OF SOLUTIONS AND SUSPENSIONS.

In an important branch of physical chemistry there is a need for a series of terms to be applied with reference to the viscosity of solutions and suspensions

Although this subject is not connected with the main substance of the preceding Report, it was considered at the meeting of the Joint Committee on Rheology, that certain recommendations should be brought forward, which would do away with the term "viscosity" for quantities not having the dimensions of a viscosity.

Let η_0 be the viscosity of the pure solvent, η that of the solution or suspension (both at the same well defined temperature). The concentration c of the suspension from the physical point of view should be given in grams solute (suspended material) per cm^3 solution. In practical work it is usually defined as grams solute per 100 mL. In all cases the unit should therefore be specified: the likewise applied to derived quantities in which the concentration plays a part.

Although physical interpretations are more conveniently based on concentrations measured per unit volume, attention must be drawn to the fact that quantities having the concentration in the denominator will become independent of the unit of volume, when the concentration is given in grams solute per gram of solution.

The dimensionless ratio

$$\eta_r = \eta/\eta_0$$

should be called relative viscosity.

The dimensionless quantity

$$\eta_s = \eta_r - 1 = \frac{\eta - \eta_0}{\eta_0}$$

should be called: relative increase of viscosity.

$$\eta_c = \frac{\eta_s}{c} = \frac{\eta_r - 1}{c} = \frac{\eta - \eta_0}{c\eta_0}$$

should be called solute viscosity factor. If no mistake is to be feared, the word "solute" can be dropped (the same remark applied to subsequent definitions). It is a quantity having the dimensions of the reciprocal concentration; hence if the latter is not given in grams per gram, it is necessary to state the units in which the concentration is measured. And one should write

$$\eta_c \left(\frac{\mathrm{cm}^3}{\mathrm{gram}}\right), \qquad \text{or,} \qquad \eta_c \left(\frac{\mathrm{dL}}{\mathrm{gram}}\right),$$

as the case may be. The density of the solution should be stated in case this is different from that of the pure solvent.

The molar solute viscosity factor is defined by

$$\eta_M = M\eta_c \left(\frac{\text{cm}^3}{\text{mol}}, \text{or} \frac{\text{dL}}{\text{mol}}\right),$$

M being the molar weight.

The chain-equivalent solute viscosity is defined by $\eta_E = E\eta_c$ cm^3/chain-equivalent, or dL/chain equivalent), where "chain-equivalent" stands for repetition unit of the polymer and E is its molecular weight.

If desired the limiting values of η_c, η_M and η_E for zero contration can be indicated by

$$[\eta_c], \qquad [\eta_M], \qquad [\eta_E]$$

respectively, giving the units as before. A proper designation would be to add "limiting" or "intrinsic" to the names given above, and the term "intrinsic viscosity factor" can be used for shortness when no ambiguity is to be feared concerning the units to which it refers.